

Physics-informed e-processes for damage detection under a lifetime false-alarm budget in continuously monitored structures

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Structured abstract

Purpose – Damage thresholds for continuously monitored structures are usually set as if one test were run, so the chance of a false alarm over a monitoring life can approach one. This paper develops a rule with a stated false-alarm budget for monitoring of unspecified length and tests how well the budget survives finite commissioning.

Design/methodology/approach – PHASE (Physics-informed Hedged Anytime-valid Sequential Evidence) predicts temperature and load effects on natural frequencies, scores the remainder only in directions stiffness loss can produce (the damage cone), and accumulates it in a betting e-process. We use 4,000 simulated frame records, sixteen comparators and variants, and Z24 bridge data.

Findings – The budget is a conditional theorem; its commissioning-based calibration is only empirical, and the provable variant has little power. With one year of commissioning, PHASE raised no false alarm in 300 undamaged records at one year or in 300 at three (upper 95% limit 1.2%). A 5% loss at storey 3 was detected in all 150 records (median 59 days), a 2% loss in 34%, and a 5% loss after two healthy years in 66%. Commissioning of 30–60 days destroyed false-alarm control. On Z24, a reduced version stayed silent when stiffness recovered; its response to settlement depended on commissioning order.

Research limitations/implications – Only simulation validates the full framework; nuisance-subspace loss patterns are indistinguishable from temperature, and storey 1 was not localised.

Practical implications – If commissioning covers the operating environment, each structure has probability at most α of ever raising a false alarm; missed detections are not

controlled.

Originality/value – First use, to our knowledge, of betting e-processes with physics-derived damage directions in vibration-based monitoring.

Keywords – Structural health monitoring; Anytime-valid inference; E-process; Physics-informed machine learning; Digital twin; Damage detection; False alarms; Infrastructure resilience; Environmental and operational variability; Predictive maintenance

Word count – approximately 20,000 words including tables, references and appendices.

1 Introduction

Instrumented buildings and bridges no longer produce measurement campaigns; they produce streams. A modern monitoring installation identifies modal properties automatically, every few hours, indefinitely (Mishra et al., 2022). The statistical machinery that decides whether those properties indicate damage, however, is largely inherited from an era of episodic testing. A control limit is placed at three standard deviations of a reference population, or a hypothesis test is applied at the 5% level, and the resulting rule is then evaluated afresh at every epoch for years. The error rate that was nominally controlled is the error rate of a single decision. The error rate that the asset owner actually experiences is the probability that the rule ever fires while the structure is healthy, and under continuous interrogation that probability tends to one.

This is not a subtle effect. In the experiments reported below, a 3σ chart applied to the standardised residuals of the same surrogate used by PHASE raised its first false alarm after a median of two days of monitoring, and did so in every one of 300 undamaged records. A repeatedly applied Hotelling T^2 test at the 1% level behaved identically. Practitioners are of course aware that raw charts alarm too often, and one common response is to raise thresholds until alarms become rare in practice. Where that is done without a stated horizon, it renegotiates the error guarantee silently: the resulting threshold has no stated operating characteristic, it is specific to the horizon and the data over which it was tuned, and, as this paper shows for the implementations tested here, buying false-alarm control this way can extinguish the power to detect the damage the system was installed to find. This is a statement about the specific charts we implemented, not about every environmental-normalisation method; Section 2.1 discusses the closest engineering studies.

The consequences run in both directions and both are material for sustainability. Alarms that cannot be trusted are eventually not acted upon, which is the mechanism by which monitoring systems quietly stop informing maintenance decisions (Cawley, 2018); each alarm that is acted upon consumes inspection resources, crew mobilisation, access

equipment and embodied carbon. Damage detected late, conversely, forecloses the repair options that preserve embodied carbon and forces replacement. A monitoring rule with a calibrated, horizon-free error guarantee is therefore not a statistical nicety. It is a precondition for monitoring to support the life-extension decisions on which the sustainability case for instrumented infrastructure rests, and it is relevant to United Nations Sustainable Development Goals 9, 11 and 13 in the sense that it concerns the decision quality on which condition-based maintenance depends; the paper does not measure sustainability outcomes.

The statistical tools required to fix this have matured only recently. Safe anytime-valid inference (SAVI) replaces the p -value with the e-value, a non-negative statistic whose expectation under the null is at most one, and replaces the fixed-sample test with a non-negative martingale whose value is interpretable as accumulated evidence (Ramdas et al., 2023; Grünwald et al., 2024; Vovk and Wang, 2021). Ville’s inequality bounds the probability that such a process ever exceeds $1/\alpha$ by α , whatever the stopping rule, whatever the horizon, and whether or not the analyst peeks. These methods have been taken up in clinical trials and online experimentation but, to our knowledge, not in vibration-based SHM, which is precisely a setting of unbounded continuous monitoring under optional stopping by a human operator. Sequential testing is not new to SHM: residual-based sequential probability ratio tests exist (Kopsaftopoulos and Fassois, 2015). What is new here is the error criterion (a time-uniform budget for the probability of ever alarming), the nonparametric conditional-mean null, and the physics-derived alternative; Section 2.3 sets out the differences.

Transplanting the machinery is not, however, sufficient, because SHM’s difficulty is not primarily multiplicity. It is that environmental and operational variation (EOV) moves modal properties by more than damage does (Sohn, 2007; Peeters and De Roeck, 2001). In the simulations reported here, seasonal and diurnal temperature variation moves the fundamental frequency by roughly $\pm 2\%$ while a 5% stiffness loss at one storey moves it by 0.49%. Any monitor is therefore only as good as its model of the nuisance, and a betting process built on a poorly specified residual stream will accumulate evidence against the

null for reasons that have nothing to do with damage. This is where physics could earn its place, and this paper examines four ways in which it can, tested to different degrees (Table 13 and Sections 4 and 5), rather than the one usually recognised.

The method developed here, PHASE (Physics-informed Hedged Anytime-valid Sequential Evidence) and sketched in Figure 1, uses first-order eigenvalue perturbation of the structure’s finite element model to: (i) constrain the EOV surrogate to $m + 2$ free parameters, with the temperature response tied across modes by an exact identity, rather than the $5m$ parameters of an unconstrained regression; (ii) restrict the alternative hypothesis to a polyhedral cone of physically admissible directions, since stiffness loss can only reduce frequencies; (iii) determine which stiffness-loss patterns are identifiable relative to the nuisance at all, since temperature and live load reach the modal set through specific directions and any damage pattern whose signature lies in the span of those directions is confounded with a nuisance change as a matter of structure rather than of statistical power; and (iv) protect identifiable damage from being absorbed by the online adaptation of the nuisance model.

Contributions. The paper makes the following contributions.

1. It formulates damage detection under continuous monitoring as an anytime-valid decision problem and gives an alarm rule, PHASE, whose family-wise false-alarm probability over an unbounded monitoring life is at most α *conditionally* on a bound for the expected score (Proposition 3). It then states precisely what is and is not established for the implemented commissioning-based calibration of that bound: a finite-sample result with an explicit calibration-failure term (Proposition 4), and an empirical assessment of how the implemented rule behaves.
2. It derives the damage cone and the nuisance sensitivity subspace from eigenvalue perturbation, and proves an identifiability result: a stiffness-loss pattern is confounded with nuisance change if and only if its standardised signature lies in the nuisance subspace (Proposition 1). Uniform loss is always confounded; it is the only confounded pattern only under additional conditions that the frame studied here

does not meet. Identifiability relative to the nuisance is distinguished from unique localisation. This yields a per-substructure detectability map that tells an engineer which parts of a structure a given modal set can support claims about.

3. It prices surrogate misspecification explicitly. If the physics-structured surrogate leaves a residual bias δ , the evidence process is discounted at rate $\log(1 + \lambda\delta)$ per epoch and remains valid (Proposition 4); a corrected delay bound that uses the relevant mixture component shows how the discount trades against detection speed (Proposition 5).
4. It evaluates the method on real measurements from the Z24 bridge progressive damage test. Only a reduced version of the method can be run there. The data support the directional discrimination and the dependence of the monitor on its commissioning data; chronological and dependence-preserving re-analyses show that the timing comparison with the omnibus variant depends on how commissioning epochs are ordered.
5. It reports a controlled simulation study over 4,000 simulated records (3,300 two-year and 700 four-year) against sixteen comparators and variants spanning current SHM practice, classical quickest detection and distribution-free anytime-valid testing, with interval estimates for all rates, a late-onset experiment, a finer commissioning sweep, a finite-sample-certified variant, a restart variant, ablations of the cone and of the adaptation rule, and a digital-twin error experiment, and it reports the costs of validity as well as its benefits.

Section 2 reviews the relevant literature. Section 3 formalises the monitoring problem, derives the physics and develops PHASE. Section 4 reports the simulation study, Section 5 the application to the Z24 bridge, and Section 6 discusses implications and concludes.

2 Background and related work

2.1 Vibration-based SHM and environmental variability

The statistical pattern recognition paradigm for SHM (Farrar and Worden, 2007, 2012) organises damage identification into operational evaluation, data acquisition, feature extraction and statistical modelling, and its axioms make explicit that damage detection always requires a comparison between two system states (Worden et al., 2007). Modal frequencies remain the most widely used global feature because they are cheap to identify from ambient response and are directly interpretable through the eigenvalue problem (Doebeling et al., 1998; Avci et al., 2021).

Their weakness is equally well established. Temperature changes the elastic modulus and boundary conditions; live load changes the mass; the resulting modal variation routinely exceeds that caused by structurally significant damage (Sohn et al., 1999; Sohn, 2007). Two decades of work on removing these effects have been surveyed recently (Wang et al., 2022; Han et al., 2021), and a 2025 critical review of baseline compensation, adaptation and reference-free techniques concludes that robustness to environmental and operational variability remains the principal obstacle to deployment (Rezazadeh et al., 2025). The Z24 bridge campaign, in which a bridge was monitored for a year before progressive artificial damage, remains the canonical demonstration that environmental effects and damage events must be separated before any detection claim is meaningful (Peeters and De Roeck, 2001). Responses in the literature include data normalisation, regression of features on measured covariates, latent-variable methods, and switching response-surface models for structures whose EOV behaviour is regime-dependent (Worden and Cross, 2018). What these approaches share is a focus on removing the nuisance; the decision rule applied to the cleaned residual is typically a control limit whose behaviour under repeated application is not analysed.

The closest primary engineering studies deserve a direct comparison. Deraemaeker et al. (2008) combine statistical compensation of environmental effects with monitoring of output-only vibration features, and Maes et al. (2022) evaluate regression-based and

robust-PCA-based compensation on measurements from the KW51 railway bridge across a retrofit. Related work couples regression, principal-component or cointegration models of the environment with control charts on frequencies or autoregressive coefficients (Fugate et al., 2001; Kullaa, 2003; Yan et al., 2005; Magalhães et al., 2012; Cross et al., 2011). These studies address the question of *how well the environment can be removed*, usually on measured data and with a fixed-sample or fixed-horizon threshold, and they evaluate detection and false-alarm rates empirically. PHASE does not compete with them on environmental compensation: its surrogate is deliberately simple. Its contribution is orthogonal, namely a decision layer whose false-alarm budget is stated over the whole monitoring life and whose alternative is restricted to physically admissible directions. The PCA-EOV chart and the control charts we implement (Section 4.1.3) are single, simple instances of this family, calibrated on simulated healthy campaigns; their poor showing here, in a simulation with deliberately unmodelled thermal lag, gradient and load-proxy error, says nothing about how well carefully tuned compensation behaves on other structures, and Maes et al. (2022) report that compensation can work well on real data.

2.2 Physics-informed learning and digital twins in the built environment

Physics-informed machine learning embeds governing equations, symmetries or known operators into learning systems, improving data efficiency and extrapolation (Karniadakis et al., 2021; Raissi et al., 2019); a 2025 review traces its adoption in bridge damage detection specifically, characterising it as a grey-box middle ground between physics-based and purely data-driven models (Mammeri et al., 2025). In civil and construction engineering the same impulse appears under the banner of digital twins, in which a calibrated model of the asset is maintained alongside the physical structure and updated from monitoring data (Jiang et al., 2021; Opoku et al., 2021; Ferrari and Willcox, 2024). Hybrid schemes that combine a physics-based model with machine learning for damage detection have

shown clear advantages over purely data-driven pipelines (Ritto and Rochinha, 2021). Bibliometric surveys of the built-environment literature confirm both the speed of uptake and its fragmentation: applications remain largely task-specific and weakly integrated into whole-asset frameworks (Wang et al., 2026), and the sustainability case for these technologies is usually argued at the level of energy and carbon rather than of decision quality (Bibri and Huang, 2025).

Almost all of the physics-informed machine-learning work cited above uses physics to improve *estimation*: a better surrogate, a better reduced-order model, better extrapolation. In SHM there is an established line that uses physics for inference: the statistical subspace-based approach builds damage residuals from model sensitivities and, for temperature, rejects the nuisance subspace by projection (Basseville et al., 2004; Balmès et al., 2009). The present paper is closest to that line in spirit, and its nuisance subspace and detectability notion are of the same kind; what it adds is a sign-restricted alternative, an identifiability-aware adaptation rule and a time-uniform error budget for unbounded monitoring. The role of physics in constraining *inference* – which alternatives are admissible, which are identifiable, and what may be learned online without destroying the detector – is therefore less explored in combination with sequential error control than in estimation, not new in itself.

2.3 Sequential monitoring and anytime-valid inference

Sequential change detection has a long history, from the sequential probability ratio test (Wald, 1945) to the CUSUM (Page, 1954) and its optimality properties (Lorden, 1971; Moustakides, 1986; Pollak, 1985). These procedures answer different questions, and the differences matter for what a false-alarm guarantee means (Table 1).

SPRT. Wald’s test decides between two specified hypotheses about the data distribution with prescribed type I and type II error probabilities; those probabilities refer to one test run until it stops. Early work applied sequential probability ratio tests to damage classification (Sohn et al., 2003), and a residual-based SPRT framework for vibration monitoring

(Kopsaftopoulos and Fassois, 2015) applies this logic to the residuals of a vibration model and supports decisions among pre-specified damage hypotheses. Restarting the test after each decision does not by itself bound the probability of ever alarming over an unbounded campaign.

CUSUM and Shiryaev–Roberts. These minimise worst-case detection delay subject to a lower bound on the *average run length* to false alarm, and they require the pre- and post-change distributions to be known or accurately estimated. An average-run-length constraint controls a mean, not a probability: if the false-alarm time is roughly geometric with mean γ , the probability of at least one false alarm within a horizon h is about h/γ for small h/γ and tends to one thereafter. In SHM practice the pre-change distribution is precisely what is uncertain, because it is the output of a fitted EOVS model applied to a non-stationary environment, and thresholds are usually set by simulation (as in our oracle comparators).

E-detectors. E-detectors (Shin et al., 2023) are nonparametric change detectors built from e-processes. Like the CUSUM family they sum evidence over candidate change points and their false-alarm guarantee is an average-run-length bound under a composite null; they therefore detect changes quickly without a known post-change law, but they do not provide the time-uniform bound $\mathbb{P}_{H_0}(\tau < \infty) \leq \alpha$ that PHASE targets.

Conformal test martingales. A conformal test martingale (Vovk, 2021) tests whether the whole observation sequence is exchangeable. Ville’s inequality bounds the probability that it ever exceeds $1/\alpha$ *under exchangeability*. Its null is therefore a statement about the joint distribution of the sequence rather than about the structure being in its reference condition: any non-exchangeability, such as a seasonal drift in a healthy structure’s score, violates the null even though nothing has been damaged.

Safe anytime-valid inference offers the last of these criteria. An e-value is a non-negative statistic with null expectation at most one; a test martingale is a running product of conditional e-values; Ville’s inequality then bounds the probability that the martingale ever exceeds $1/\alpha$ by α (Ramdas et al., 2023; Grünwald et al., 2024). Betting formulations construct these processes as the wealth of a gambler placing predictable bets (Waudby-

Table 1: Error criteria and assumptions of the sequential procedures discussed in Section 2.3, and where PHASE sits among them.

Procedure	Null / model assumed	Error criterion	Consequence for monitoring
SPRT	Two specified densities (simple or parametric)	Type I and II error of one test run to a decision	Per-run control; restarts need their own budget
CUSUM, Shiryaev–Roberts	Known or estimated pre- and post-change laws	Average run length to false alarm; minimax delay	Mean, not probability, of false alarm; probability within a horizon grows with the horizon
E-detector	Composite null with e-process under it	Average run length lower bound, non-parametric	Quick change detection without post-change law; no time-uniform probability bound
Conformal test martingale	Exchangeable sequence	$\mathbb{P}(\sup M_t \geq 1/\alpha) \leq \alpha$ under exchangeability	Valid only if calibration and monitoring scores are exchangeable
PHASE (this paper)	Conditional mean of a bounded score $\leq \mu_0 + \delta$	$\mathbb{P}_{H_0}(\tau < \infty) \leq \alpha$ for any dependence and stopping rule	Lifetime budget, conditional on the mean bound; slower than oracle CUSUM

Smith and Ramdas, 2024; Kelly, 1956), time-uniform confidence sequences arise from the same machinery (Howard et al., 2021), and e-detectors extend it to change detection (Shin et al., 2023). E-values also combine gracefully: arbitrary dependent e-values may be merged by averaging (Vovk and Wang, 2021), and false discovery rate control across many hypotheses follows directly (Wang and Ramdas, 2022), which matters when a structure is divided into many monitored zones.

2.4 Research gap

Three gaps motivate this work. First, SHM decision rules are evaluated at fixed sample sizes but deployed continuously, and the literature reports few if any measurements of the family-wise false-alarm probability actually experienced over a monitoring life. Second, physics-informed SHM has concentrated on the nuisance model, leaving the admissibility and identifiability of damage alternatives largely implicit. Third, anytime-valid inference has not, to our knowledge, been connected to a domain in which its central assumption

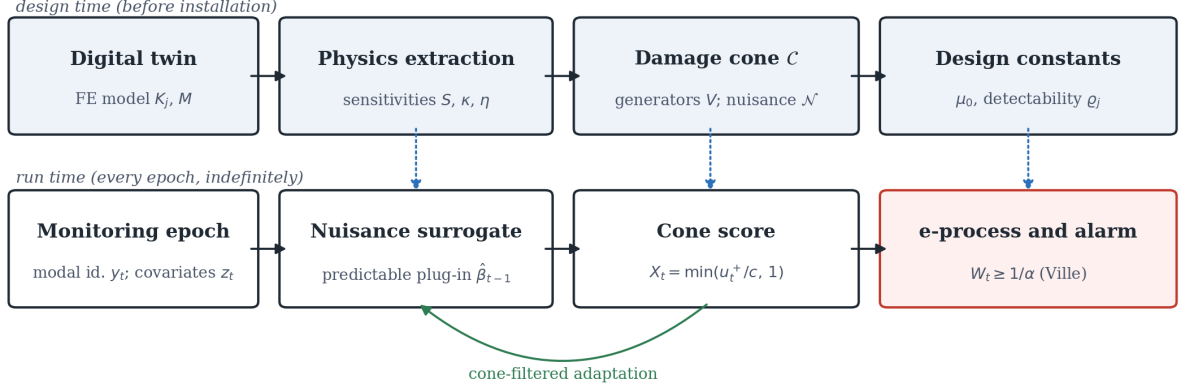


Figure 1: The PHASE monitor. Quantities in the upper row are computed once, before installation, from the digital twin: the modal sensitivities, the damage cone and the nuisance subspace, and the design constants μ_0 and the detectability map. The lower row runs at every interrogation epoch for the life of the installation. The feedback path is the identifiability-aware adaptation of Section 3.9, which allows the nuisance model to track drift while withholding residual components consistent with identifiable damage.

– unbounded monitoring with operator-driven stopping – is not an idealisation but a description of ordinary practice. Sequential tests for SHM do exist (Sohn et al., 2003; Kopsaftopoulos and Fassois, 2015), as do compensation-plus-monitoring studies (Deraemaeker et al., 2008; Maes et al., 2022); we are not aware of earlier work that combines a nonparametric e-process with a physics-derived alternative cone and an identifiability-aware adaptation rule, and we make no wider claim of priority.

3 Problem formulation and the PHASE monitor

3.1 Monitoring protocol and the anytime-valid decision problem

A structure is interrogated at epochs $t = 1, 2, \dots$. At each epoch an automated operational modal analysis returns the first m identified natural frequencies, collected as $y_t = \log \hat{f}_t \in \mathbb{R}^m$, together with measured covariates z_t comprising ambient temperature deviation ΔT_t and a live-load proxy q_t . Let $\mathcal{F}_t = \sigma(y_1, z_1, \dots, y_t, z_t)$.

A commissioning window $t \leq t_0$ is used to calibrate the monitor; monitoring proper runs for $t > t_0$ without a specified end. A notation and acronym list is given in Appendix A. Write H_0 for the hypothesis that the structure remains in its reference condition

throughout. A monitor is a stopping time τ , the alarm time. The quantity of operational interest is

$$\text{FWFA}(\alpha) = \mathbb{P}_{H_0}(\tau < \infty), \quad (1)$$

the probability that the monitor *ever* alarms on a healthy structure. We require $\text{FWFA} \leq \alpha$ for a horizon that is neither fixed nor known in advance. This is a strictly stronger requirement than per-epoch error control, and strictly stronger than average-run-length control, since it constrains the entire monitoring life rather than a rate. It concerns the first alarm only: after an alarm and an investigation the monitor is restarted, and a restart needs its own share of the budget (Section 6.1).

In words. An *e-process* is a running tally of evidence against “the structure is unchanged”, built like a gambler’s wealth: at each epoch the monitor bets a fraction of its current wealth that the score will exceed its healthy average. If the structure really is unchanged, the wealth cannot grow on average, so reaching $1/\alpha$ times the starting wealth is very unlikely, and that holds no matter when, or how often, one looks.

3.2 Physics: sensitivities, the damage cone, and the nuisance subspace

Let M be the mass matrix and $K(\theta) = \sum_{j=1}^n \theta_j K_j$ the stiffness matrix, with K_j the contribution of substructure j and θ_j a multiplier equal to one in the reference state. Writing $\lambda_i = \omega_i^2$ for the eigenvalues and ϕ_i for the mode shapes of $K\phi = \lambda M\phi$, first-order perturbation gives

$$\frac{\partial \lambda_i}{\partial \theta_j} = \frac{\phi_i^\top K_j \phi_i}{\phi_i^\top M \phi_i}, \quad S_{ij} := \frac{\phi_i^\top K_j \phi_i}{\lambda_i \phi_i^\top M \phi_i} \geq 0. \quad (2)$$

Since $\log \omega_i = \frac{1}{2} \log \lambda_i$, a stiffness loss $\rho_j = 1 - \theta_j \geq 0$ at substructure j shifts the log-frequencies by

$$\Delta \log \omega = -\frac{1}{2} S \rho + O(\|\rho\|^2). \quad (3)$$

Two structural facts follow and are used throughout. First, S is elementwise non-negative: stiffness loss can only lower frequencies. Second, because $K(\theta)$ is homogeneous of degree

one in θ , a uniform scaling $\theta = (1 - \rho)\mathbf{1}$ scales all eigenvalues equally, whence

$$\sum_j S_{ij} = 1 \quad \text{for every mode } i. \quad (4)$$

A uniform change in elastic modulus, which is how temperature principally acts, therefore shifts every log-frequency by the same amount. Writing $\kappa = S\mathbf{1} = \mathbf{1}$ for the uniform-modulus sensitivity and $\eta_i = (\sum_{k \in \mathcal{L}} \phi_{ik}^2 m_k) / (\phi_i^\top M \phi_i)$ for the sensitivity to relative added mass on loaded floors $\mathcal{L}$, the nuisance reaches the modal set only through the two directions κ and η , whatever the functional form of the underlying EOV law.

Let $D = \text{diag}(\sigma_1, \dots, \sigma_m)$ hold the identification scatter of each log-frequency and define standardised quantities $\tilde{d}_j = -D^{-1}S e_j$. The *damage cone* is the finitely generated convex cone

$$\mathcal{C} = \left\{ -D^{-1}S\rho : \rho \in \mathbb{R}_+^n \right\} = \text{cone}\{\tilde{d}_1, \dots, \tilde{d}_n\}, \quad (5)$$

which is pointed because all its generators lie in the negative orthant. The *nuisance sensitivity subspace* is $\mathcal{N} = \text{span}\{D^{-1}\kappa, D^{-1}\eta\}$, with orthogonal projector $P_\perp = I - \Pi_{\mathcal{N}}$. *In words.* The damage cone collects every frequency-change pattern that some non-negative loss of stiffness could produce; a change outside it (all frequencies rising, for example) is not damage of that kind. The nuisance subspace collects the frequency-change patterns that temperature and live load can produce. Damage whose signature lies inside that subspace looks, to a frequency-based monitor, like weather or occupancy.

Proposition 1 (Identifiability relative to the nuisance). *Let $A = D^{-1}S$. (a) To first order, a stiffness-loss pattern ρ produces a modal signature that is indistinguishable from some combination of a uniform modulus change and a live-load change if and only if $A\rho \in \mathcal{N}$. These patterns form a linear subspace $\mathcal{K} = \{\rho : P_\perp A\rho = 0\}$ of dimension $n - \text{rank}(P_\perp A) \geq n - (m - \dim \mathcal{N})$. (b) Uniform stiffness loss $\rho \propto \mathbf{1}$ always lies in $\mathcal{K}$. (c) If A has full column rank n and the range of A contains no nuisance direction other than $D^{-1}\kappa$ (in particular if temperature is the only nuisance), then $\mathcal{K} = \text{span}\{\mathbf{1}\}$. Conversely, $\mathcal{K} \supsetneq \text{span}\{\mathbf{1}\}$ whenever $\ker A \neq \{0\}$ (in particular whenever $n > m$) or whenever $\mathcal{N}$ contains a direction in the range of A that is not parallel to $D^{-1}\kappa$. (d) The*

identifiable part of the signature of ρ is $P_{\perp}A\rho$. Two patterns ρ, ρ' can be told apart from frequencies, to first order, only if $P_{\perp}A(\rho - \rho') \neq 0$; hence at most $\text{rank}(P_{\perp}A) \leq m - \dim \mathcal{N}$ independent combinations of damage can be resolved, and a non-zero identifiable fraction of each substructure is necessary but not sufficient for unique localisation among n substructures.

The proof is given in the Appendix. The practical content is considerable: no monitor, however sophisticated, can separate a pattern in $\mathcal{K}$, and in particular a global loss of stiffness, from a warming of the structure using modal frequencies alone, and the ratio

$$\varrho_j = \frac{\|P_{\perp}\tilde{d}_j\|}{\|\tilde{d}_j\|} \in [0, 1] \quad (6)$$

gives a design-time *detectability map* over the structure: the fraction of substructure j 's signature that survives confounding with the nuisance. Two cautions follow from parts (c) and (d). First, the statement that thermal confounding occurs only for uniform loss is not true in general: for the eight-storey frame of Section 4.1 ($n = 8$, $m = 6$, $\dim \mathcal{N} = 2$) the confounded subspace has dimension four, and equal stiffness loss at storeys 1 and 8 is confounded exactly (the projected signatures are antiparallel, with cosine -1.000 and a numerical residual below 10^{-13}). Second, $\varrho_j > 0$ is identifiability *relative to the nuisance*, not localisability: it says that substructure j 's signature is not wholly absorbed by temperature and load, but says nothing about whether it can be distinguished from the signature of substructure k . The pairwise cosine $\langle P_{\perp}\tilde{d}_j, P_{\perp}\tilde{d}_k \rangle / (\|P_{\perp}\tilde{d}_j\| \|P_{\perp}\tilde{d}_k\|)$ measures how alike the identifiable signatures of single-storey damage are; with only $m - \dim \mathcal{N} = 4$ resolvable directions for eight storeys, the eight identifiable signatures cannot be linearly independent. Section 4.2.5 shows that the map is informative about detection only weakly and about localisation only partly, for reasons that are themselves instructive.

3.3 Physics-structured nuisance surrogate with sequential plug-in

The surrogate predicts reference-condition log-frequencies from covariates. Physics fixes its structure:

$$g_\beta(z_t)_i = a_i + b_T \kappa_i \Delta T_t + b_q \eta_i q_t, \quad \beta = (a_1, \dots, a_m, b_T, b_q), \quad (7)$$

with κ_i and η_i supplied by the digital twin through (2). The model has $m + 2$ free scalars; the temperature response is a single scalar b_T shared across all modes, an exact consequence of (4). (The load coefficient b_q is unrelated to the score-saturation scale c of equation (9).) The comparator used in the ablations is an unconstrained per-mode quadratic regression on $[1, \Delta T, \Delta T^2, q, \Delta T q]$ with $5m$ free parameters.

Coefficients are updated recursively with forgetting factor ν , and crucially the prediction at epoch t uses $\hat{\beta}_{t-1}$, estimated from data up to $t - 1$ only. The standardised residual

$$r_t = D^{-1}(y_t - g_{\hat{\beta}_{t-1}}(z_t)) \quad (8)$$

is thus a *sequential plug-in* residual, and any statistic formed from $\hat{\beta}_{t-1}$ is $\mathcal{F}_{t-1}$ -measurable. This predictability, not the accuracy of $\hat{\beta}$, is what preserves the martingale property in Section 3.5.

3.4 Polyhedral approximation of the damage cone and the score

For a closed convex cone, $\sup_{v \in \mathcal{C}, \|v\| \leq 1} \langle v, r \rangle = \|\Pi_{\mathcal{C}}(r)\|$. Rather than solving a non-negative least-squares projection at every epoch, PHASE fixes a finite set $V = \{v_1, \dots, v_N\}$ of unit vectors in $\mathcal{C}$ – the normalised generators together with normalised pairwise combinations – and scores

$$u_t = \max_{k \leq N} \langle v_k, r_t \rangle, \quad X_t = \min(\max(u_t, 0)/c, 1) \in [0, 1]. \quad (9)$$

Evaluating (9) is a single $N \times m$ matrix–vector product, which matters for edge deployment on a gateway alongside the sensor array.

Proposition 2 (Polyhedral approximation). *For every r , $\max_k \langle v_k, r \rangle \leq \|\Pi_{\mathcal{C}}(r)\|$, and if V has covering radius γ in $\mathcal{C} \cap \mathbb{S}^{m-1}$ then $\max_k \langle v_k, r \rangle \geq \|\Pi_{\mathcal{C}}(r)\| - 2 \sin(\gamma/2) \|r\|$. The approximation is monotone: refining V cannot decrease the score.*

The null hypothesis is stated directly in terms of the score,

$$H_0 : \quad \mathbb{E}[X_t \mid \mathcal{F}_{t-1}] \leq \mu_0 \quad \text{for all } t, \quad (10)$$

where $\mu_0 = \mathbb{E}_{\varepsilon \sim N(0, I_m)} [\min(\max_k \langle v_k, \varepsilon \rangle^+ / c, 1)]$ is a design-time constant obtained once by Monte Carlo from the digital twin. Stating the null this way is deliberate: it makes explicit that the surrogate’s task is to render (10) true in the reference condition, and Section 3.6 prices the failure to do so rather than assuming it away. In words: given everything observed up to the previous epoch, the score of an undamaged structure is on average no larger than the constant μ_0 that the digital twin predicts for ideal, correctly modelled noise. The assumption is about conditional averages of a bounded score; it needs no distributional form and no independence, but it is an assumption about the future, and Section 3.6 explains what it demands of commissioning.

3.5 Betting e-process and the alarm rule

For $\lambda \in [0, 1/\mu_0)$ define the wealth process $W_0^\lambda = 1$ and

$$W_t^\lambda = \prod_{s=t_0+1}^t (1 + \lambda(X_s - \mu_0)), \quad W_t = \frac{1}{|\Lambda|} \sum_{\lambda \in \Lambda} W_t^\lambda, \quad (11)$$

mixing over a fixed geometric grid Λ so that no bet size need be tuned (Waudby-Smith and Ramdas, 2024). The alarm is

$$\tau = \inf\{t > t_0 : W_t \geq 1/\alpha\}. \quad (12)$$

Proposition 3 (Anytime validity). *Under (10), $(W_t)_{t>t_0}$ is a non-negative supermartingale with $W_{t_0} = 1$, and consequently $\mathbb{P}_{H_0}(\exists t : W_t \geq 1/\alpha) \leq \alpha$. The bound holds for arbitrary dependence in (y_t, z_t) , for any predictable bet sequence, for sequential plug-in estimation of $\hat{\beta}$, and irrespective of when or how often the operator inspects the process.*

Two features deserve emphasis. First, no pre-change distribution need be known: only the conditional mean bound (10) is required, and the score is bounded by construction, so no tail assumption enters. Proposition 3 is a conditional statement: it does not say that the bound (10) holds for the implemented surrogate, which is the subject of Section 3.6. Second, $\log W_t$ is directly interpretable as accumulated evidence in nats, so an operator may act on a rising process before it crosses the threshold without invalidating anything – the guarantee is uniform over stopping rules.

3.6 Pricing surrogate misspecification and calibrating the allowance

Real EOV laws are never captured exactly. Thermal lag, vertical temperature gradients and imperfect load proxies leave a residual bias, so that (10) holds only up to some $\delta \geq 0$. Write $m_t = \mathbb{E}[X_t | \mathcal{F}_{t-1}] - \mu_0$ for the conditional mean excess of the score, an $\mathcal{F}_{t-1}$ -measurable quantity, and let $\hat{\delta} \geq 0$ be an allowance that is fixed at the end of commissioning ($\mathcal{F}_{t_0}$ -measurable). Define the discounted component wealth $\widetilde{W}_t^\lambda = \prod_{s=t_0+1}^t \frac{1+\lambda(X_s-\mu_0)}{1+\lambda\hat{\delta}}$ and the mixture $\widetilde{W}_t = |\Lambda|^{-1} \sum_{\lambda \in \Lambda} \widetilde{W}_t^\lambda$.

Proposition 4 (Bias discount and its calibration). *(i) Known allowance. If $m_t \leq \delta$ for all $t > t_0$ and $\hat{\delta} = \delta$ is deterministic, $\widetilde{W}^\lambda$ and $\widetilde{W}$ are non-negative supermartingales and $\mathbb{P}(\exists t : \widetilde{W}_t \geq 1/\alpha) \leq \alpha$.*

(ii) Estimated allowance. For any $\mathcal{F}_{t_0}$ -measurable $\hat{\delta}$, let $\sigma = \inf\{t > t_0 : m_t > \hat{\delta}\}$ be the first epoch at which the allowance is too small. Then

$$\mathbb{P}(\exists t : \widetilde{W}_t \geq 1/\alpha) \leq \alpha + \mathbb{P}(\sigma < \infty).$$

The second term is the calibration-failure probability and belongs in the total error budget.

(iii) A finite-sample certified allowance. Let C be the n epochs of the second half of commissioning, whose scores are sequential plug-in scores. Assume representativeness: $\sup_{t>t_0} m_t \leq \bar{m}_C := n^{-1} \sum_{s \in C} m_s$ almost surely, that is, the future conditional excess never exceeds its average over commissioning. For $\varepsilon \in (0, 1)$ put $\hat{\delta}_\varepsilon = (\bar{X}_C - \mu_0 + \sqrt{\log(1/\varepsilon)/(2n)})^+$. Then $\mathbb{P}(\sigma < \infty) \leq \varepsilon$, so the alarm rule with threshold $1/(\alpha - \varepsilon)$ and allowance $\hat{\delta}_\varepsilon$ has FWFA $\leq \alpha$, whatever the dependence between epochs.

Proposition 4 separates three statements that are easily conflated. Part (i) is the conditional theorem: validity given a uniform bound on the conditional excess. Part (ii) shows that a data-dependent allowance converts that theorem into a bound with an additional term, the probability that the allowance is ever too small. Part (iii) shows what it takes to control that term: a concentration inequality for the commissioning average that does not need independence (Azuma–Hoeffding, since the scores lie in $[0, 1]$ and are plug-in), and an assumption linking the future to the commissioning period. That assumption, representativeness, is stronger than coverage of the operating envelope. It bounds the *largest* future conditional excess by the *average* commissioning excess. If the bias of the surrogate varies with season or load, the maximum exceeds the average even when commissioning has spanned the whole cycle, so representativeness fails: in the simulated generator the 90-day block means of the excess oscillate by about ± 0.04 around an average close to zero (Section 4.2.2). Representativeness therefore describes a regime in which the bias is nearly constant in time, not the general seasonal case, and part (iii) shows what a certified calibration needs and why it is conservative; it does not justify the implemented rule. Replacing the commissioning average by the largest of several block averages, with a union bound over the blocks, would accommodate a slowly varying bias, but it still requires the future per-epoch excess not to exceed the largest block average, which the same diagnostics show to be violated at the epoch scale. No procedure can verify the assumption from commissioning data alone, and Section 4.2.6 shows what happens when commissioning does not span the cycle.

What the implemented rule establishes. PHASE as implemented estimates the allowance as $\hat{\delta} = (\bar{X} - \mu_0 + 2\hat{\text{se}})^+$ from the second half of the commissioning window, with $\hat{\text{se}}$ the sample standard error of the plug-in scores, and discounts the evidence at $\log(1 + \lambda\hat{\delta})$ nats per epoch. This is a calibration heuristic. The factor two corresponds to a one-sided error of about 2.3% only if the scores were independent and the sample mean Gaussian, the standard error ignores serial dependence (the scores are only weakly autocorrelated at lag one, but their mean varies with the season, Section 4.2.2), the error is not subtracted from the alarm budget, and representativeness is untested. Proposition 4 therefore does *not* establish that the implemented procedure has $\text{FWFA} \leq \alpha$. What it establishes is the conditional statement of part (i) and the form that any calibration guarantee must take; whether the implemented rule meets the budget is an empirical question, answered for the simulated scenarios in Section 4.2.2 and for short commissioning windows in Section 4.2.6. To show the price of a provable statement we also evaluate PHASE-C, which uses part (iii) with $\varepsilon = \alpha/2$ and the threshold $2/\alpha$; its guarantee is conditional on representativeness and nothing else.

Throughout the paper, “valid” therefore means *conditionally valid* when it refers to Propositions 3 and 4, and “calibrated” or “observed” when it refers to the false-alarm frequencies of the implemented rule. Estimation uncertainty, serial dependence and later environmental variation enter the guarantee only through $\mathbb{P}(\sigma < \infty)$ and the representativeness assumption.

This converts model quality into a currency: a surrogate that leaves half the bias buys back half the discount, and the mechanism gives a precise sense in which better physics is worth more than better statistics.

3.7 Detection delay

Proposition 5 (Delay bound). *Fix $\lambda \in \Lambda$ and a change time $t^* \geq t_0$. Let $L_t^\lambda = \log \widetilde{W}_t^\lambda$ be the log-wealth of the discounted component with bet λ , and suppose that for all $s > t^*$,*

almost surely,

$$\mathbb{E}[\log(1 + \lambda(X_s - \mu_0)) \mid \mathcal{F}_{s-1}] - \log(1 + \lambda\hat{\delta}) \geq \gamma_\lambda > 0.$$

Let $b = \log(1/\alpha) + \log|\Lambda|$, let $B_\lambda = \max\{0, \log(1 + \lambda(1 - \mu_0)) - \log(1 + \lambda\hat{\delta})\}$ bound the increments of L^λ from above, and write $x^- = \max(-x, 0)$. Then the alarm time τ of the mixture rule satisfies

$$\mathbb{E}[(\tau - t^*)^+ \mid \mathcal{F}_{t^*}] \leq \frac{b + B_\lambda + (L_{t^*}^\lambda)^-}{\gamma_\lambda}. \quad (13)$$

If the drift condition holds for several λ , the bound holds for each and the smallest may be taken.

The bound has the expected shape: delay grows logarithmically in $1/\alpha$ and in the size of the bet grid, and inversely in the evidence growth rate γ_λ , from which the bias discount subtracts directly ($\gamma_\lambda = g(\lambda) - \log(1 + \lambda\hat{\delta})$ when $g(\lambda)$ lower-bounds the undiscounted log-growth). Two points deserve emphasis. The drift condition concerns one component, so the starting-wealth penalty must use that component's own log-wealth $L_{t^*}^\lambda$, not the total mixture wealth; the two can differ greatly, because after a long healthy period the components with large bets have lost most of their wealth while the mixture is carried by components with small bets. And the penalty $(L_{t^*}^\lambda)^-$ is the quantitative form of “evidence lost during a long healthy period”: it enters the delay one-for-one at rate $1/\gamma_\lambda$. Section 4.2.7 tests this with late damage onset, and compares the bound with realised delays. Restricting the alternative to the cone raises $g(\lambda)$ by removing the noise that an omnibus statistic must accumulate over all directions, which is why the cone ablation in Section 4 loses so much power.

3.8 Restarts: protecting against evidence lost in long healthy periods

The penalty $(L_{t^*}^\lambda)^-$ in (13) has a practical consequence. During a long healthy period the evidence process drifts downwards: the allowance $\hat{\delta}$ is subtracted at every epoch, and the logarithm of a bet has a negative drift (Jensen) even when the conditional mean excess is zero. The components with large bets, which are the ones that detect damage quickly, lose the most. A change that starts late is then detected slowly, however large it is. The remedy is to hedge over the start time of the evidence process as well as over the size of the bet.

Proposition 6 (Start-time mixture). *Let $s_0 < s_1 < \dots$ be fixed start times with $s_0 = t_0$ and let $\pi_k \geq 0$ with $\sum_k \pi_k \leq 1$. Let $\widetilde{W}_t^{(k)} = 1$ for $t \leq s_k$ and, for $t > s_k$, let $\widetilde{W}_t^{(k)}$ be the mixture over Λ of $\prod_{s_k < u \leq t} (1 + \lambda(X_u - \mu_0)) / (1 + \lambda\hat{\delta})$. Under the conditions of Proposition 4(i), $W_t^R = \sum_k \pi_k \widetilde{W}_t^{(k)}$ is a non-negative supermartingale with initial value $\sum_k \pi_k \leq 1$, so $\mathbb{P}(\exists t : W_t^R \geq 1/\alpha) \leq \alpha$. If the change occurs at t^* and $k^* = \max\{k : s_k \leq t^*\}$, then for every λ for which the drift condition of Proposition 5 holds for the component started at s_{k^*} , the bound (13) holds for the alarm time of W^R with b replaced by $b + \log(1/\pi_{k^*})$ and $L_{t^*}^\lambda$ replaced by the log-wealth of that component, started at s_{k^*} and observed up to t^* .*

PHASE-R, evaluated below, starts a new evidence process every 90 days (360 epochs) with weights $\pi_k = 1/((k+1)(k+2))$, which sum to one. The price is the extra $\log(1/\pi_k) \approx 2\log(k+2)$ nats in the numerator of (13), which grows slowly with the age of the installation; the gain is that the deficit term is bounded by what accumulates in at most 90 days. The construction is in the spirit of change-point maximisation in e-detectors (Shin et al., 2023) but keeps the time-uniform probability bound, because the components are averaged and not maximised. Like Proposition 3, validity is conditional on the bound on the conditional mean excess, and the same representativeness caveat applies.

3.9 Identifiability-aware adaptation

Online adaptation of a nuisance model in a damage detector is dangerous: with forgetting factor ν , the surrogate absorbs a persistent step within roughly $1/(1 - \nu)$ epochs, and a detector whose reference model chases the damage will eventually go quiet. Freezing the surrogate after commissioning avoids this but forfeits tracking of genuine long-term drift, and in preliminary experiments a naive evidence-gated freeze produced a positive feedback loop in which a frozen surrogate generated ever-larger residuals.

Physics resolves the tension. Let $\tilde{V}$ denote the generators of the cone projected onto $\mathcal{N}^\perp$ and normalised, and let $w_t = \max_k \langle \tilde{v}_k, r_t \rangle^+ \tilde{v}_{k^*}$ be the identifiable damage-consistent component of the residual. The surrogate is updated towards the target

$$y_t - D(r_t - \text{clip}(r_t - w_t, \pm\varsigma)), \quad (14)$$

that is, the identifiable damage-consistent component is withheld from adaptation while everything else, Huber-clipped at ς , is retained. Because the nuisance acts within $\mathcal{N}$ and the withheld component lies in $\mathcal{N}^\perp$, the surrogate remains free to learn the entire temperature and live-load response – which it must, since temperature acts inside the damage cone – while identifiable damage cannot be absorbed. The rule is a fixed function of r_t and so preserves predictability.

3.10 Localisation and multiple zones

The generator attaining the maximum in (9) carries a support over substructures. Accumulating $L_j += \text{supp}_j(v_{k^*})(X_t - \mu_0)^+$ over the monitoring period yields a localisation index whose argmax names the implicated substructure. The index is a heuristic accumulator, not a calibrated estimator, and carries no error guarantee of its own. For structures partitioned into zones, a separate e-process per zone can be run and the zone-level processes merged by averaging, which is valid under arbitrary dependence (Vovk and Wang, 2021); if zone-level claims are wanted, e-values support false discovery rate control directly

Algorithm 1 PHASE monitoring cycle (one epoch)

- 1: **Design time:** S, κ, η from the twin; generators $V, \tilde{V}$; projector $P_\perp$; μ_0 by Monte Carlo
 - 2: **Commissioning:** fit $\hat{\beta}$; estimate D ; estimate $\hat{\delta}$ on the second half of the commissioning epochs, scored by the sequential plug-in rule
 - 3: **for** $t > t_0$ **do**
 - 4: $r_t \leftarrow D^{-1}(y_t - g_{\hat{\beta}_{t-1}}(z_t))$ $\triangleright$ sequential plug-in
 - 5: $u_t \leftarrow \max_k \langle v_k, r_t \rangle$; $X_t \leftarrow \min(u_t^+ / c, 1)$
 - 6: $\log W_t^\lambda \leftarrow \log W_{t-1}^\lambda + \log(1 + \lambda(X_t - \mu_0)) - \log(1 + \lambda\hat{\delta})$ for $\lambda \in \Lambda$
 - 7: **if** $\log \text{meanexp}_\lambda \log W_t^\lambda \geq \log(1/\alpha)$ **then** raise alarm; report localisation index
 - 8: update $\hat{\beta}$ towards the cone-filtered target (14)
 - 9: **end for**
-

(Wang and Ramdas, 2022), avoiding the conservatism of a Bonferroni correction across zones.

4 Simulation study

4.1 Experimental design

4.1.1 Digital twin and data generation

The test structure, drawn in Figure 2, is an eight-storey shear frame with storey mass 2.5×10^5 kg and storey stiffness 6.0×10^8 N/m, giving reference frequencies of 1.44, 4.27, 6.95, 9.40, 11.52 and 13.26 Hz for the $m = 6$ identified modes. Epochs are six hours apart, corresponding to automated modal identification four times per day, and records span two years (2,920 epochs) with a one-year commissioning window unless stated otherwise.

Environmental forcing follows a northern-Vietnam monsoon calendar: an annual temperature cycle of $24 \pm 7^\circ\text{C}$, a diurnal cycle of amplitude 4°C , and autocorrelated weather noise, giving an ambient range of roughly $10\text{--}39^\circ\text{C}$. Three features are deliberately *not* available to the monitor. Storey temperatures follow ambient through first-order thermal lags with storey-dependent time constants and a vertical gradient spanning $\pm 15\%$; the live load acts on floors 2–7 with an occupancy and weekday cycle and is observed only through a proxy with 10% multiplicative error; and modal identification noise is Student-

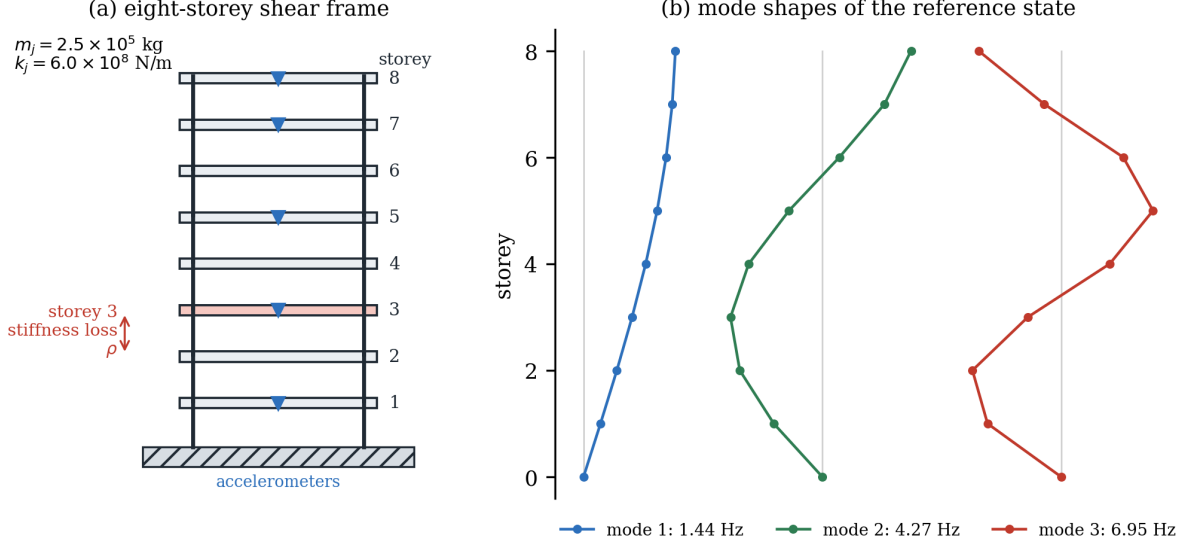


Figure 2: The simulated structure. (a) Eight-storey shear frame with lumped storey masses and storey stiffnesses; damage is introduced as a multiplicative stiffness loss at a single storey, storey 3 unless stated otherwise. (b) Mode shapes and natural frequencies of the reference state, from which the sensitivity matrix S of equation (2) is computed.

t with five degrees of freedom, scaled to 0.60–1.20% relative standard deviation across modes. The monitor sees only single-sensor ambient temperature and the load proxy, so its surrogate is misspecified in exactly the way field surrogates are. All settings of the generator, the monitors and the comparators are listed in Appendix B.

Modulus depends on temperature with $\beta_T = 2.5 \times 10^{-3} \text{ } ^\circ\text{C}^{-1}$. Frequencies are computed exactly by solving the generalised eigenproblem at each epoch, not by the perturbation formula used by the monitor, so the first-order model is genuinely approximate from the monitor’s point of view.

4.1.2 Scenarios

Seven non-healthy scenarios are simulated, each with onset at epoch 1,825 (year 1.25): abrupt stiffness loss of 2%, 5% and 10% at storey 3; a gradual loss reaching 6% at storey 3 over 60 days; abrupt 5% loss at storeys 1 and 6, chosen for their contrasting identifiable fractions; and a benign 3% *stiffening* of all storeys, representing temporary propping during façade works, which must not trigger a damage alarm. Undamaged records are simulated over two years and over four years to test behaviour beyond a calibration

horizon. Three further groups were added to answer questions about commissioning, onset time and the digital twin: a finer sweep of the commissioning window; records of four years with damage onset late, at epoch 4,745 (year 3.25), after two years of healthy monitoring; and records in which the monitor’s structural model carries errors in stiffness and mass.

4.1.3 Comparators

All comparators consume the same residual stream where possible, so that differences reflect the decision rule rather than the features. Variant labels are used identically in text, tables and figures. **PHASE** is the full method with the commissioning-estimated allowance $\hat{\delta}$ of Section 3.6. **PHASE-ND** omits the allowance ($\hat{\delta} = 0$; “no discount”). **PHASE-C** uses the finite-sample certified allowance of Proposition 4(iii) with $\varepsilon = \alpha/2$ and the alarm threshold $2/\alpha$. **PHASE- Ω** replaces the cone score with the omnibus norm $\|r_t\|$, retaining all other machinery. **PHASE-BB** replaces the physics-structured surrogate with the unconstrained per-mode regression. **PHASE-E** additionally suspends betting when covariates fall outside the commissioning envelope.

Four comparators represent current SHM practice. The 3σ **chart** alarms when any $|r_{it}| > 3$. The **calibrated chart** uses a threshold set at the 99th percentile of the healthy maximum over a one-year monitoring horizon, obtained from 150 independent healthy simulations. The **Mahalanobis novelty index** is the classical outlier statistic $\sqrt{r_t^\top \hat{\Sigma}^{-1} r_t}$ with a threshold calibrated the same way rather than taken from a χ^2 table. The **PCA-EOV chart** implements one standard data-normalisation strategy directly: the leading two principal components of the commissioning log-frequency matrix, which are dominated by environmental response, are projected out and the norm in the retained subspace is charted, again with a simulation-calibrated threshold. The **repeated T^2 test** applies a χ_m^2 test at the 1% level every epoch. These are single, simple implementations, not the best that tuned compensation can achieve.

Three comparators represent classical sequential detection, all applied to the same cone score and all with thresholds calibrated to a one-year horizon on 150 healthy simulations:

CUSUM, the **Shiryaev–Roberts** procedure, and an **EWMA** chart. These are oracles in a strong sense, since they require 150 replicated realisations of the true healthy process, which no real deployment possesses.

Finally, two **conformal test martingales** are anytime-valid comparators that obtain their guarantee differently from PHASE. Cone scores from a *fixed* calibration set serve as the reference; each monitoring epoch yields a smoothed conformal p -value p_t (the rank of X_t among the calibration scores, ties broken at random), and a mixture of power martingales $\int_0^1 \prod_s \epsilon p_s^{\epsilon-1} d\epsilon$, approximated on a grid of twelve values of ϵ in $[0.05, 0.95]$, accumulates evidence with the same Ville alarm at $1/\alpha$ (Vovk, 2021). The first (**CTM**) uses the second half of the commissioning window as the calibration set, as PHASE does for $\hat{\delta}$; the second (**CTM-F**) uses scores from the whole commissioning window, which spans a full seasonal cycle. Both use the identical score and alarm rule, so the comparison isolates one question: whether validity should rest on exchangeability of the calibration sample or, as in PHASE, on a conditional mean bound with an explicit misspecification discount.

The validity of these comparators rests on a specific assumption, which we state because it determines when their guarantee should be expected to hold. The p_t are uniform and independent, and the product is a test martingale, if the monitoring scores are exchangeable with the calibration scores and the calibration set is treated as fixed. The intended reference condition is therefore that the healthy structure’s score sequence is stationary and free of serial dependence, so that commissioning and monitoring scores come from one distribution. The simulated healthy structure does not meet that condition in two ways that real structures also fail: the scores are weakly serially dependent and vary with the season (thermal lag and weather persistence; Section 4.2.2 reports their autocorrelation), and the surrogate that generates them keeps adapting. A failure of the guarantee for these comparators is therefore a failure of their stated null in this setting and not an error in Ville’s inequality.

4.1.4 Metrics and accounting

We report the family-wise false-alarm probability (1) over one- and three-year monitoring horizons; detection rate and detection delay after onset; pre-onset false alarms in damage scenarios; and localisation accuracy. Rates are reported as counts with exact (Clopper–Pearson) 95% confidence intervals, which are informative even when no event is observed: zero events in n records give an upper limit of $1 - 0.025^{1/n}$, that is 1.2% for $n = 300$ and 0.6% for $n = 600$. These intervals describe sampling variability across simulated structures drawn from one generator; they are not statements about other generators or about real structures.

The remaining quantities are defined as follows, because the treatment of records that never alarm, or that alarm too early, changes the numbers.

- *Pre-onset alarm.* An alarm at an epoch before damage onset. Such a record is a false alarm in the damage scenario; it is counted in the false-alarm column and is *not* counted as a detection. Where the number of such records is non-zero it is stated.
- *Detection and detection rate.* A record is detected if its first alarm falls at or after onset and within the remaining nine months of the record (1,095 epochs after onset, 4×274 days). The detection rate is the number of detected records divided by all records of the scenario.
- *Undetected record.* No alarm in the whole record. Undetected records are right-censored at the end of the monitoring window.
- *Detection delay.* Days from onset to first alarm. Medians and interquartile ranges (IQR) are computed *among detected records only*, so they are conditional on detection; when the detection rate is below one half the unconditional median is undefined (the median record is censored). A conditional median should be read together with the detection rate.
- *Localisation.* The localisation index is read at the PHASE alarm epoch. Accuracy

Table 2: Accounting of simulated records. All records use independent seeds. Two-year records have 2,920 epochs; four-year records 5,840.

Group	Purpose	Records
Calibration	healthy, two-year; thresholds of calibrated chart, CUSUM and other comparators	150
Healthy, two-year	Table 4 one-year horizon; Table 6 reference	300
Healthy, four-year	Table 4 three-year horizon	300
Damage scenarios	seven scenarios $\times$ 150, two-year (Table 6)	1,050
Commissioning sweep	seven window lengths $\times$ (100 healthy + 100 damaged), two-year (Table 8)	1,400
Late onset	four scenarios $\times$ 100, four-year (Table 9)	400
Twin error	two error levels $\times$ (100 healthy + 100 damaged), two-year (Table 11)	400
Total	3,300 two-year and 700 four-year records	4,000

is the fraction of *detected* records in which the index names the damaged storey; its denominator is the number of detected records, stated beside the percentage.

Table 2 gives the accounting of the simulated records. The first four groups total 1,800 records; the last three, a finer commissioning sweep, a late-onset experiment and a twin-error experiment, answer questions about commissioning, onset time and the fidelity of the digital twin.

4.2 Results

4.2.1 Validity of the first-order physics

Table 3 compares the exact log-frequency shift from re-solving the eigenproblem with the first-order prediction $-\frac{1}{2}S\rho$. At 2% stiffness loss the maximum relative error across modes is below 1.9%, rising to 9.3% at 10% loss. The perturbation basis on which the cone and the sensitivity structure rest is therefore accurate over the damage range of interest, with the approximation degrading gracefully and conservatively.

4.2.2 False-alarm control under continuous monitoring

Table 4 reports the probability that each monitor ever alarms on an undamaged structure, with exact 95% intervals. The 3σ chart and the repeated T^2 test alarmed in every record

Table 3: Accuracy of the first-order sensitivity model against exact eigen-solution. Relative error is the maximum across the six identified modes.

Stiffness loss	Storey 1	Storey 3	Storey 6
2%	1.9%	1.7%	1.8%
5%	4.6%	4.2%	4.6%
10%	9.3%	8.2%	9.2%

(interval 0.988–1.000), at median elapsed times of 2.0 and 3.5 days respectively. This is the expected consequence of applying a fixed-sample rule 1,460 times, and it is the behaviour that drives alarm fatigue in deployed systems.

PHASE and its variants PHASE-E, PHASE- Ω , PHASE-BB and PHASE-C raised no false alarm in 300 records at either horizon. An observed zero in 300 records is compatible with a true rate of up to 1.2%, which is above the nominal 1%; the two horizons use different groups of records, and pooling them gives 0/600 records with an upper limit of 0.6%, which describes the pooled set and not either horizon. The upper limits quoted are those of two-sided 95% intervals (the one-sided 95% limit for zero events in 300 is 1.0%). The result is therefore consistent with the target and does not prove it, and it describes the generator used here. It also describes the implemented rule, not the theorem: Section 3.6 explains why the commissioning-based allowance carries no guarantee of its own.

The more interesting comparisons are with the methods that do control error to some degree. Given 150 replications of the true healthy process, the oracle-calibrated chart holds to 0.007 over the horizon for which it was tuned, but its false-alarm probability then grows as monitoring continues, reaching 0.023 at three years (Figure 3). The Mahalanobis novelty index behaves identically, and the PCA-EOV chart – our implementation of one standard data-normalisation strategy – is worse at both horizons, at 0.017 and 0.040 (the three-year interval, 0.021–0.069, excludes 0.01). The distinction is not a matter of a better-chosen threshold: for the Shewhart-type charts a threshold calibrated for a horizon carries no guarantee beyond it, whereas a supermartingale threshold is valid at all horizons simultaneously, given its conditional assumption. (The oracle CUSUM and Shiryaev–Roberts procedures, calibrated in the same way, stayed at 0/300 at three years in this generator, but nothing in their construction guarantees that.)

Table 4: Family-wise false-alarm probability on undamaged structures, with exact 95% confidence intervals in brackets. One-year results from 300 two-year records; three-year results from 300 four-year records. The calibrated chart and CUSUM thresholds were tuned for a one-year monitoring horizon. PHASE-R is the restart variant of Section 3.8.

Monitor	1-year horizon	3-year horizon
<i>anytime-valid by construction (conditionally)</i>		
PHASE	0.000 [0.000, 0.012]	0.000 [0.000, 0.012]
PHASE-C	0.000 [0.000, 0.012]	0.000 [0.000, 0.012]
PHASE-R	0.030 [0.014, 0.056]	0.023 [0.009, 0.047]
PHASE-E	0.000 [0.000, 0.012]	0.000 [0.000, 0.012]
PHASE- Ω	0.000 [0.000, 0.012]	0.000 [0.000, 0.012]
PHASE-BB	0.000 [0.000, 0.012]	0.000 [0.000, 0.012]
PHASE-ND	0.003 [0.000, 0.018]	0.017 [0.005, 0.038]
CTM	0.033 [0.016, 0.060]	0.030 [0.014, 0.056]
CTM-F	0.000 [0.000, 0.012]	0.000 [0.000, 0.012]
<i>sequential detection, horizon-calibrated oracles</i>		
CUSUM	0.003 [0.000, 0.018]	0.000 [0.000, 0.012]
Shiryayev–Roberts	0.003 [0.000, 0.018]	0.000 [0.000, 0.012]
EWMA	0.003 [0.000, 0.018]	0.020 [0.007, 0.043]
<i>current SHM practice</i>		
Calibrated chart (oracle)	0.007 [0.001, 0.024]	0.023 [0.009, 0.047]
Mahalanobis novelty index	0.007 [0.001, 0.024]	0.023 [0.009, 0.047]
PCA-EOV chart	0.017 [0.005, 0.038]	0.040 [0.021, 0.069]
3σ chart	1.000 [0.988, 1.000]	1.000 [0.988, 1.000]
Repeated T^2 test	1.000 [0.988, 1.000]	1.000 [0.988, 1.000]
Nominal level $\alpha = 0.01$. Zero events in 300 records give an upper limit of 0.012.		

The conformal test martingale needs a careful reading. With the second half of the commissioning window as calibration set (CTM) it exceeded the nominal level, at 0.033 and 0.030 (intervals 0.016–0.060 and 0.014–0.056). With a calibration set drawn from the whole commissioning window (CTM-F) there was no false alarm in 300 records at either horizon. The excess of CTM is therefore not evidence that conformal calibration cannot control error. It reflects a calibration set that covers only half a seasonal cycle and is not exchangeable with the monitoring scores, which is the assumption its guarantee needs (Section 4.1.3). The same lesson applies to PHASE: its theorem needs the allowance to represent the future, and Section 4.2.6 shows what happens when commissioning does not span the cycle. What distinguishes the two is the form of the assumption (exchangeability of the score sequence for CTM; a conditional mean bound with an explicit allowance for PHASE), and, as Table 6 shows, power: CTM-F detected 3% of 2% losses where PHASE

detected 34%, and was slower at 5% loss.

The value of the allowance is visible in the same table. Without it (PHASE-ND), the false-alarm probability was 0.003 at one year and 0.017 at three (interval 0.005–0.038), as the unmodelled thermal lag and gradient accumulated evidence at a slow but non-zero rate. With $\hat{\delta}$ estimated from commissioning data (mean 0.020, standard deviation 0.011 across the 150 calibration records) no false alarm occurred in 600 healthy records totalling 1,200 structure-years of monitoring (300 one-year and 300 three-year monitoring periods). The certified allowance of PHASE-C (mean 0.059) also gave no false alarm, at a large cost in power (Table 6). This zero should not be read as support for Proposition 4(iii): the PHASE-C allowance is essentially the Hoeffding margin (about 0.060 for 730 scores and $\varepsilon = 0.005$) and is larger than the seasonal excess in most records, so representativeness is not what protects it.

How well do the assumptions hold in the simulation? The conditional mean excess m_t cannot be observed, but its time averages can be estimated from the healthy records. Over the 600 healthy records the median monitoring-period mean of $X_t - \mu_0$ was -0.003 , against a median allowance of 0.020, and it exceeded the allowance in 8 records (1.3%). The lag-one autocorrelation of the score was weak (median 0.04; 10th to 90th percentile 0.01–0.06). The 90-day block means, however, exceeded the allowance in at least one block in 94.5% of the records (the largest of 4 or 12 blocks per record; 34.0% of the 4,800 individual blocks exceeded it). Scores are also computed under Student- t identification noise, which lowers the null mean of the score by about 0.018 relative to the Gaussian value μ_0 used by the monitor, a slack of the same size as the allowance and specific to this generator. The uniform bound of Proposition 4(i), which is sufficient for validity, is therefore not literally satisfied at the scale of a season: the bias of the surrogate varies seasonally and diurnally. The discounted process still never alarmed, because over a whole campaign the negative drift of the log-wealth (the allowance plus the Jensen term) outweighs short stretches of positive excess. The absence of false alarms is thus not evidence that the theorem’s condition holds; it shows slack in the allowance

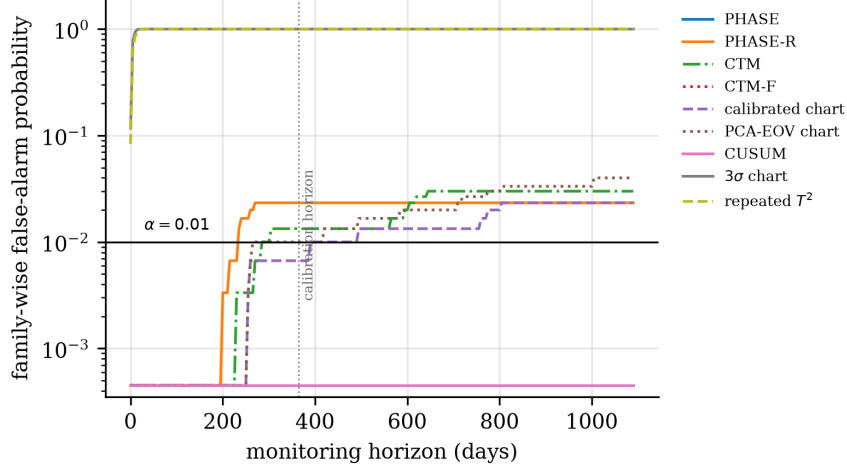


Figure 3: Family-wise false-alarm probability as a function of monitoring horizon, from 300 undamaged four-year records. Curves for PHASE and the oracle CUSUM lie on the axis floor (no false alarm observed). The calibrated chart controls error only up to the horizon for which it was tuned.

for this generator, and the restart variant (Section 4.2.7), whose components see only a few months, shows that the slack is limited.

4.2.3 Detection delay, localisation, and the cost of validity

Table 5 reports PHASE detection performance with intervals and the handling of undetected and early-alarm records defined in Section 4.1.4, and Table 6 compares variants and comparators. PHASE detected 5% stiffness loss at storey 3 in 150 of 150 records (95% interval 0.97–1.00) with a median delay of 59 days (interquartile range 51–69 days), and 10% loss with a median delay of 26 days (22–28). At 2% loss – a signature amounting to roughly 0.2% frequency change against 0.6–1.2% identification scatter – detection within the remaining nine months occurred in 51 of 150 records (34%; interval 0.26–0.42). The median delay of 156 days (128–186) is conditional on detection: the 99 undetected records are censored at nine months, and because fewer than half of the records were detected the median over all records is undefined, so the typical 2% loss is *not* detected within nine months. Gradual degradation reaching 6% over 60 days was detected in every record, with a median delay of 80 days from onset of degradation. No PHASE record in any damage scenario alarmed before onset.

Table 5: PHASE detection and localisation by scenario, 150 records each. Detection means a first alarm at or after onset within the remaining nine months; the rate is detected over all records, with an exact 95% interval. Delay is the median in days (interquartile range) among detected records only. Pre-onset: records with an alarm before onset. Undetected: records with no alarm. Localisation: records in which the index read at the alarm epoch names the damaged storey, over detected records.

Scenario	Detected	Rate [95% CI]	Delay, days (IQR)	Pre-onset	Undetected	Localised
2% storey 3	51/150	0.34 [0.26, 0.42]	156 (128–186)	0	99	46/51
5% storey 3	150/150	1.00 [0.97, 1.00]	59 (51–69)	0	0	150/150
10% storey 3	150/150	1.00 [0.97, 1.00]	26 (22–28)	0	0	150/150
Gradual 6% storey 3	150/150	1.00 [0.97, 1.00]	80 (74–88)	0	0	147/150
5% storey 1	150/150	1.00 [0.97, 1.00]	38 (33–43)	0	0	0/150
5% storey 6	150/150	1.00 [0.97, 1.00]	62 (55–70)	0	0	149/150
Benign 3% stiffening	0/150	0.00 [0.00, 0.02]	—	0	150	—

Table 6: Detection rate / median detection delay in days among detected records, 150 records per scenario. PHASE-R is the restart variant; its observed false-alarm probability exceeds the nominal level (Table 4), so its speed is not comparable on equal terms. PCA-EOV chart and Mahalanobis index: detection rate ≤ 0.02 in every damage scenario. Median delays are conditional on detection (Section 4.1.4).

Scenario	PHASE	PHASE-R	PHASE-C	PHASE- Ω	PHASE-BB	CTM	CTM-F	CUSUM
2% storey 3	0.34 / 156	0.94 / 67	0.00 / —	0.00 / —	0.43 / 142	0.79 / 121	0.03 / 187	0.91 / 96
5% storey 3	1.00 / 59	1.00 / 10	0.95 / 116	0.29 / 157	1.00 / 57	1.00 / 44	1.00 / 71	1.00 / 26
10% storey 3	1.00 / 26	1.00 / 3	1.00 / 39	0.99 / 34	1.00 / 24	1.00 / 16	1.00 / 22	1.00 / 10
Gradual 6%	1.00 / 80	1.00 / 48	0.99 / 134	0.44 / 179	1.00 / 77	1.00 / 66	1.00 / 92	1.00 / 53
5% storey 1	1.00 / 38	1.00 / 6	1.00 / 64	0.60 / 108	1.00 / 37	1.00 / 28	1.00 / 39	1.00 / 16
5% storey 6	1.00 / 62	1.00 / 11	0.93 / 126	0.22 / 140	1.00 / 60	1.00 / 48	1.00 / 73	1.00 / 28
Benign stiffening	0.00 / —	0.00 / —	0.00 / —	1.00 / 11	0.00 / —	0.00 / —	0.00 / —	0.00 / —

The results describe a frontier rather than a winner, and the costs of validity should be stated plainly. Every method that detects faster than PHASE pays for it somewhere. The oracle CUSUM and Shiryaev–Roberts procedures detect roughly twice as fast throughout and hold their error rate, but they are calibrated from 150 replications of the true healthy process and their guarantee is an average run length rather than a probability over the monitoring life; neither is available to a real deployment. The EWMA chart is faster still, at a median of 11 days for 5% damage, but its false-alarm probability reaches 0.020 by three years and it detects only 17% of 2% losses. The conformal martingale with the half-window calibration set detects 5% damage in 45 days against PHASE’s 59, and 79% of 2% losses against PHASE’s 34%, but at three times the nominal false-alarm rate; with the full-window calibration set (CTM-F) it holds the level but is slower than PHASE at 5% loss (71 against 59 days) and for gradual loss, similar at 10% loss, and detects only

3% of 2% losses. PHASE, PHASE-BB, PHASE-E and CTM-F all combine zero observed false alarms at both horizons with detection of every loss of 5% or more. PHASE is therefore not unique in this respect. PHASE-BB and PHASE-E are variants of the same framework, and CTM-F consumes the same cone score. PHASE’s distinguishing features are its stated conditional guarantee, its higher power for small losses than CTM-F, and, through the cone, its discrimination of benign change; it has no advantage over PHASE-BB once a full year of commissioning is available.

The discount is part of that price: undiscounted PHASE (PHASE-ND) detects 5% loss in 43 days rather than 59, and 95% of 2% losses rather than 34%, while drifting above the nominal level. The certified variant pays far more: PHASE-C detected 143 of 150 losses of 5% at a median of 116 days and none of the 2% losses. Proposition 5 makes the exchange rate explicit. Evaluating the bound of Proposition 5 with the realised log-growth rate of each record as γ_λ (a plug-in illustration, not a test of the drift condition: it uses the growth realised over the whole record, a post hoc minimum over λ and detected records only, so it cannot fail by construction and shows only the scale of the bound) and the realised component log-wealth at onset, the bound held in all 150 records of each of the four scenarios examined (5% and 10% at storey 3, 5% at storeys 1 and 6), with a median bound about 1.6 times the realised delay; for 5% loss at storey 3 the median bound was 94 days against an observed median of 59.

At the other end of the frontier sit the two data-normalisation comparators. The PCA-EOV chart and the Mahalanobis index detected at most 2% of damaged records in every scenario while also failing to control error beyond their calibration horizon – the worst of both, and a caution against reading a quiet chart as a well-behaved monitor, although these are single implementations whose thresholds were calibrated on simulated healthy campaigns.

Two further observations concern the stiffening scenario. The EWMA, CUSUM, Shiryaev–Roberts and conformal comparators all remained silent there, but this is inherited rather than earned: each consumes the physics-derived cone score, and the directional restriction is what discards the change. Only PHASE- Ω , which sees the omnibus norm,

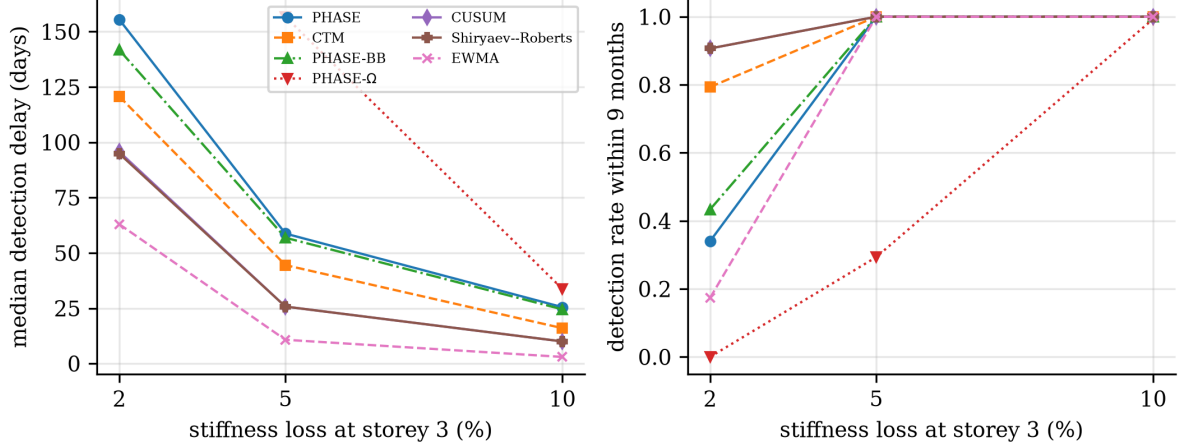


Figure 4: Median detection delay among detected records (left) and detection rate within the remaining nine months (right) against damage severity at storey 3. The oracle CUSUM is faster throughout; the cone-free ablation PHASE-Ω loses most of the power.

alarms. The physics, not the sequential machinery, supplies the discrimination.

The horizon-calibrated chart illustrates the opposite failure. Its threshold had to reach 24.3 standardised units to survive the EOVS-contaminated residual stream, and it consequently detected 5% damage in none of 150 records and 10% damage in 1.3%. Controlling false alarms by raising a threshold, rather than by accumulating evidence, sacrifices essentially all of the detection capability the monitoring system was installed to provide. Figure 4 summarises the severity–delay relationship for all variants.

4.2.4 Discriminating benign change: the value of the damage cone

The clearest single result concerns the stiffening scenario, in which temporary propping raises all storey stiffnesses by 3%. This is a real change in the structure, and any omnibus change detector should and does find it: PHASE-Ω alarmed in all 150 records, at a median of 11 days. PHASE alarmed in none, because the change lies outside the damage cone – frequencies rose, and stiffness loss cannot raise frequencies. The oracle CUSUM, which in our implementation consumes the same cone score, also correctly stayed silent, which underlines that the discriminating ingredient is the physics-derived restriction of the alternative and not the sequential machinery. That discrimination is a consequence of the sign assumption: a damage mechanism that raised some frequencies would be treated

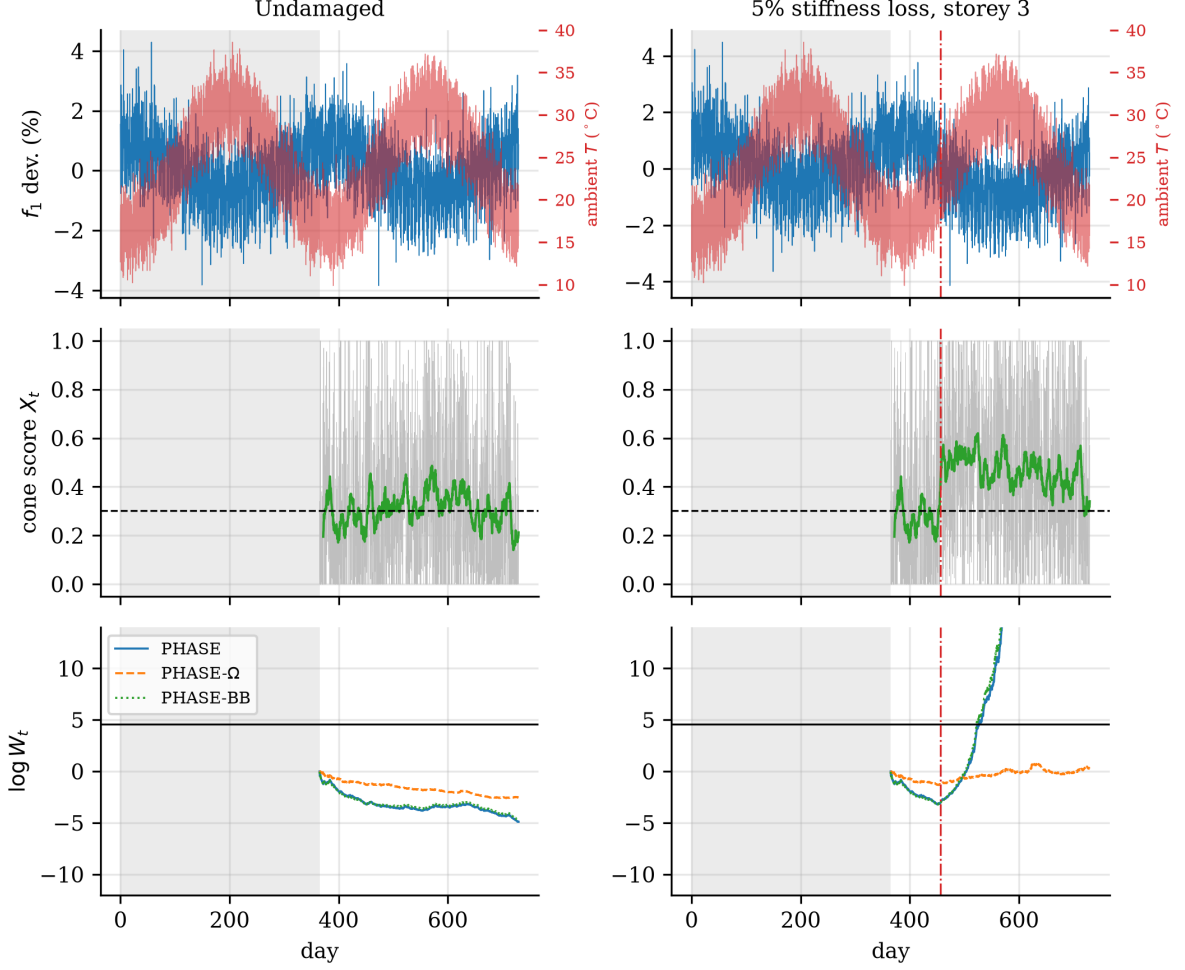


Figure 5: One undamaged (left) and one damaged (right) record. Top: fundamental-frequency deviation (blue) against ambient temperature (red); environmental variation exceeds the damage signature. Middle: cone score with 7-day mean and null mean μ_0 (dashed). Bottom: accumulated log-evidence for PHASE and two ablations, with the alarm threshold $\log(1/\alpha)$. Shading marks commissioning; the dash-dotted line marks damage onset.

in the same way (Section 5.2).

The cone contributes power as well as specificity. Removing it while holding everything else fixed reduced the detection rate for 5% damage from 1.00 to 0.29 and roughly tripled the median delay (Table 6). An omnibus statistic must accumulate noise over all m directions; the cone restricts attention to the ones damage can actually produce.

4.2.5 The detectability map: what physics can and cannot promise

Figure 6 shows the identifiable fraction (6) for each storey. It ranges from 0.245 at storey 1 to 0.588 at storey 6: between 41% and 76% of each storey’s modal signature is confounded

with the nuisance directions and is unavailable for inference, however long one monitors.

The map’s empirical content is more subtle than we anticipated, and the discrepancy is worth reporting. Detection delay did *not* follow the identifiable fraction: 5% loss at storey 1, whose identifiable fraction is lowest, was detected fastest (median 38 days) because its *total* signature norm is largest (57.5 versus 42.2 standardised units for storey 3). The resolution is that the surrogate absorbs the confounded component only over its adaptation timescale, roughly 2,000 epochs here, which is longer than the detection delays observed. Over practical horizons detection is therefore governed by the full signature; the identifiable fraction governs the asymptotic regime.

Localisation does not follow the map either, and the obvious explanation of it does not survive scrutiny. With the index read at the alarm epoch, damage at storeys 3 and 6 was localised correctly in 150 of 150 and 149 of 150 detected records, and 2% damage at storey 3 in 46 of 51. Damage at storey 1 was localised correctly in none of 150: it was attributed to storey 3 in 61 records, to storey 6 in 56, to storey 7 in 18, to storey 2 in 9 and to storey 4 in 6. One might attribute this to the storeys whose identifiable directions best match the residual once the confounded component is removed. That is not what the geometry says: in the idealised limit in which the nuisance-aligned component of the storey-1 signature is fully absorbed, the cone generator nearest to the remaining signature is storey 1’s own, and the pairwise cosines of the identifiable signatures of storey 1 with storeys 3 and 6 are low (0.27 and 0.17). The failure must therefore come from finite-sample noise, from partial absorption by the adaptive surrogate, and from the index itself. The last point can be checked: on the 300 undamaged records the index, read at the end of the record, named storey 6 in 99, storey 8 in 95, storey 7 in 46, storey 3 in 44, storey 2 in 13 and storey 4 in 3 records, and never storeys 1 or 5. The index has a baseline preference for some storeys independent of damage, and storey 6 is its most frequent healthy winner; the correct localisation at storey 6 should be read with that in mind, and a debiased index would be needed before localisation claims could be made. The identifiable signatures of storeys 1 and 8 are exactly antiparallel (cosine -1.000), a case of the confounding in Proposition 1(c). Detection and localisation are distinct capabilities, and identifiability

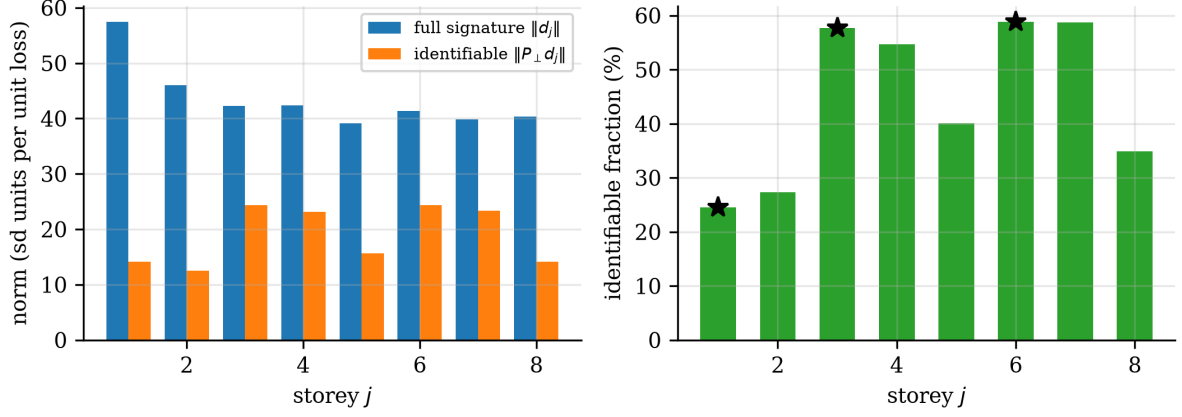


Figure 6: Physics-derived detectability map. Left: full signature norm and the part surviving projection out of the nuisance subspace. Right: identifiable fraction by storey; stars mark the storeys used in the damage scenarios.

relative to the nuisance, which the map reports, predicts neither of them reliably in this study. For asset managers this is still actionable, but in a weaker form than the map alone suggests: before any sensor is installed the map says which substructures lose most of their signature to the nuisance, not which a proposed modal set can localise.

The map depends on the nuisance model assumed. Table 7 recomputes it for additional nuisance directions: a vertical temperature gradient, a change in the support condition at storey 1 (or an equivalent roof or equipment mass change, which gives the same result here), and both. Each added direction removes one of the $m - \dim \mathcal{N}$ resolvable dimensions and lowers the identifiable fractions; a support-condition nuisance makes storeys 1 and 8 entirely confounded and storey 2 nearly so. Effects that act as a uniform change of modulus or mass, as humidity may, fall inside the existing directions and add nothing; effects with spatial structure add a direction each. Wind, in this frequency-only setting, would matter to the extent that it changes stiffness, mass or boundary conditions. The detectability map should therefore be recomputed for the nuisance set judged plausible for the asset, and its entries read as upper bounds.

4.2.6 Commissioning window: a structural prerequisite

Table 8 reports a sweep over commissioning length. The results are unforgiving. With 30 or 60 days of commissioning the monitor is invalid: false-alarm probabilities of 0.24 and

Table 7: Identifiable fraction ϱ_j by storey for alternative nuisance models ($m = 6$ modes, eight storeys). $\dim \mathcal{N}$: dimension of the nuisance subspace; “left”: $m - \dim \mathcal{N}$ resolvable directions.

Nuisance model	$\dim \mathcal{N}$	left	1	2	3	4	5	6	7	8	min
temperature only (κ)	1	5	0.28	0.54	0.58	0.55	0.40	0.60	0.60	0.63	0.276
temperature + live load (κ, η) [baseline]	2	4	0.25	0.27	0.58	0.55	0.40	0.59	0.59	0.35	0.245
+ vertical gradient	3	3	0.21	0.27	0.26	0.52	0.36	0.56	0.43	0.30	0.210
+ support condition (or roof mass)	3	3	0.00	0.20	0.56	0.55	0.38	0.58	0.58	0.00	0.000
+ gradient + support	4	2	0.00	0.13	0.23	0.51	0.36	0.51	0.28	0.00	0.000

Table 8: Commissioning-window sweep: false-alarm probability on undamaged records and detection rate of 5% loss at storey 3, from 100 records per cell (a zero in 100 records has an upper limit of 0.036). Coverage is the fraction of monitoring epochs whose covariates fall inside the commissioning envelope. $\hat{\delta}$: mean PHASE allowance.

Days	Coverage	$\hat{\delta}$	PHASE		PHASE-C		PHASE-BB		PHASE-E
			FWFA	Det.	FWFA	Det.	FWFA	Det.	FWFA
30	0.44	0.076	0.24	0.23	0.00	0.00	0.73	0.05	0.11
60	0.45	0.074	0.25	0.09	0.00	0.00	0.63	0.05	0.16
120	0.75	0.119	0.01	0.00	0.00	0.00	0.09	0.00	0.04
180	0.97	0.121	0.00	0.00	0.00	0.00	0.00	0.00	0.00
240	0.99	0.092	0.00	0.01	0.00	0.00	0.00	0.00	0.00
300	0.98	0.057	0.00	0.60	0.00	0.00	0.00	0.69	0.00
365	0.98	0.020	0.00	1.00	0.00	0.97	0.00	1.00	0.00

0.25 for PHASE (24 and 25 of 100 records), and 0.73 and 0.63 for the black-box variant. The mechanism is extrapolation. The bias discount $\hat{\delta}$ measures misspecification inside the observed covariate envelope, and a monitor commissioned in one season spends the rest of the year outside it. At 120 days validity returns (1 of 100) but detection collapses to zero, because the enlarged $\hat{\delta}$ consumes all the evidence a 5% signal can generate. Between 180 and 240 days the range of temperature seen in commissioning covers 97–99% of the monitoring epochs and there is no false alarm, yet the allowance is still large (0.12 and 0.09) and almost nothing is detected; at 300 days detection reaches 60% and at 365 days 100% with $\hat{\delta} = 0.020$. Coverage of the covariate range is therefore necessary but not sufficient: the allowance keeps falling until the window spans the seasonal cycle, and $\hat{\delta}$ itself is the more informative diagnostic. With high coverage a large $\hat{\delta}$ makes the monitor valid but powerless, a visible failure; with low coverage, as at 30 and 60 days, an allowance of 0.07–0.08 measures only the bias inside the observed envelope and understates what the structure does outside it, and the monitor is invalid. Both diagnostics are needed.

Within this bleak picture, the physics-structured surrogate degrades markedly more gracefully than the black-box alternative – a factor of three in false-alarm probability at 30

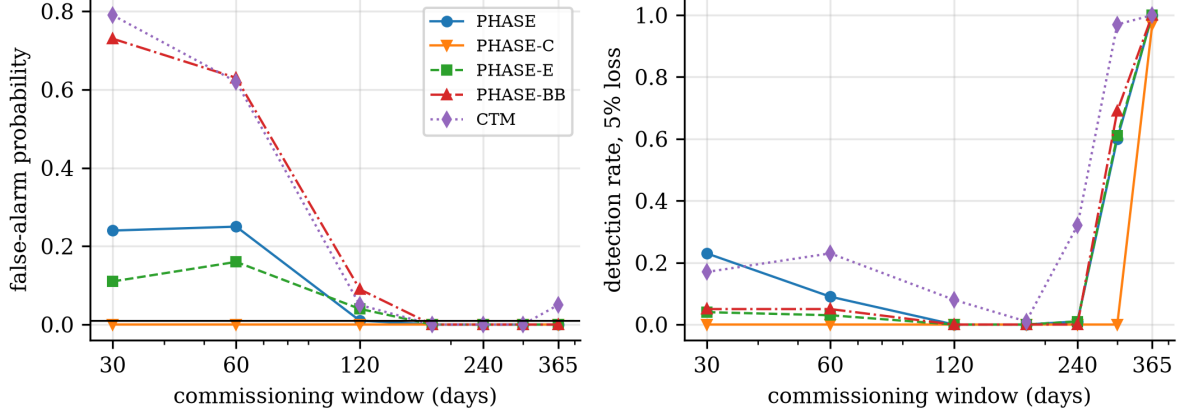


Figure 7: Effect of commissioning-window length on false-alarm probability (left) and on detection of 5% stiffness loss (right), from 100 records per point. Validity and power coexist only when commissioning spans the operational envelope.

days – which is the clearest evidence in our study for the value of physics in the nuisance model. Envelope gating, which suspends betting outside the commissioning envelope, roughly halves the false-alarm probability at 30 days but monitors only 44% of epochs and does not restore validity. At 120 days gating was worse than no gating (0.04 against 0.01). This is a negative result with a constructive implication, taken up in Section 6.

Notably, once a full seasonal cycle is available, the black-box surrogate matches the physics-structured one on both validity and delay (Table 6). With 1,460 epochs of commissioning data there is ample information to fit 30 parameters, and the advantage of physical structure is largely exhausted. Physics buys data efficiency and graceful degradation, not unconditional superiority, and we think it is more useful to say so than to select the comparison that flatters the method.

4.2.7 Late damage onset: does evidence lost in a long healthy period matter?

All the scenarios above start damage three months after commissioning. Table 9 places onset at year 3.25 of a four-year record, after two years of healthy monitoring, with the same nine months remaining. The result confirms the concern raised by the delay bound. At the median the mixture log-wealth at onset was -7.2 , so the process starts the damage period nearly seven nats in deficit. PHASE detected 5% loss at storey 3 in 66 of 100 records (against 150 of 150 with early onset), with a median delay of 203

Table 9: Late onset (epoch 4,745, year 3.25) in four-year records, 100 records per scenario, nine months remaining. Rate over all records with exact 95% interval; delay in days, median (interquartile range) among detected records. Pre-onset: alarms before onset.

Scenario	Monitor	Detected [95% CI]	Delay, days (IQR)	Pre-onset
5% storey 3	PHASE	66/100 [0.55, 0.75]	203 (162–239)	0
	PHASE-R	97/100 [0.91, 0.99]	25 (22–28)	3
	PHASE-C	0/100 [0.00, 0.03]	—	0
	CTM-F	22/100 [0.14, 0.31]	224 (198–237)	0
	CUSUM	100/100 [0.96, 1.00]	27 (25–29)	0
2% storey 3	PHASE	0/100 [0.00, 0.03]	—	0
	PHASE-R	90/100 [0.82, 0.95]	94 (84–118)	2
	PHASE-C	0/100 [0.00, 0.03]	—	0
	CTM-F	0/100 [0.00, 0.03]	—	0
	CUSUM	100/100 [0.96, 1.00]	93 (80–106)	0
10% storey 3	PHASE	100/100 [0.96, 1.00]	83 (68–103)	0
	PHASE-R	99/100 [0.94, 1.00]	9 (8–10)	1
	PHASE-C	92/100 [0.84, 0.96]	206 (186–232)	0
	CTM-F	100/100 [0.96, 1.00]	85 (75–100)	0
	CUSUM	100/100 [0.96, 1.00]	10 (9–10)	0
Benign stiffening	PHASE	0/100 [0.00, 0.03]	—	0
	PHASE-R	0/100 [0.00, 0.03]	—	4
	PHASE-C	0/100 [0.00, 0.03]	—	0
	CTM-F	0/100 [0.00, 0.03]	—	0
	CUSUM	0/100 [0.00, 0.03]	—	1

days (against 59), detected none of the 2% losses (against 34%), and needed 83 days for 10% loss (against 26). Benign stiffening still raised no alarm. PHASE-BB behaved similarly, and the certified variant PHASE-C detected no 5% loss. The oracle CUSUM, which consumes the same score and is calibrated for a one-year horizon, detected every late 5% loss at a median of 27 days and every late 2% loss at 93 days, about eight and more than three times faster than PHASE respectively (and PHASE detected no 2% loss); CTM-F detected 22% of the 5% losses and none of the 2% losses. The long healthy period therefore costs power substantially, an effect absent from the early-onset tables and one that a monitoring life of decades would make worse.

The start-time mixture of Section 3.8 removes most of this penalty. PHASE-R detected 97 of 100 late 5% losses at a median of 25 days, 90 of 100 late 2% losses, and was faster than PHASE even with early onset (10 days for 5% loss at storey 3, Table 6). It does not come free. Its observed false-alarm probability was 0.030 at one year and 0.023 at three (Table 4; the one-year interval, 0.014–0.056, excludes the nominal 0.01), and ten of

Table 10: Ablations. Detection rate / median delay in days among detected records (150 records per scenario; 100 for late onset). False alarms: records with an alarm among 300 two-year (one-year horizon) and 300 four-year (three-year horizon) healthy records. PHASE-S: sign-only score. PHASE-NA: no withholding in the adaptation. PHASE- Ω : no sign restriction.

Scenario	PHASE	PHASE-S	PHASE-NA	PHASE- Ω
2% storey 3	0.34 / 156	0.02 / 252	0.33 / 152	0.00 / —
5% storey 3	1.00 / 59	0.99 / 91	1.00 / 59	0.29 / 157
10% storey 3	1.00 / 26	1.00 / 34	1.00 / 26	0.99 / 34
Gradual 6%	1.00 / 80	1.00 / 111	1.00 / 80	0.44 / 179
5% storey 1	1.00 / 38	1.00 / 57	1.00 / 38	0.60 / 108
5% storey 6	1.00 / 62	0.99 / 95	1.00 / 63	0.22 / 140
Benign stiffening	0.00 / —	0.00 / —	0.00 / —	1.00 / 11
Late onset, 5% storey 3	0.66 / 203	0.06 / 257	0.61 / 201	0.02 / 219
False alarms, 1-year ; 3-year	0/300 ; 0/300	0/300 ; 0/300	0/300 ; 0/300	0/300 ; 0/300

the 400 late-onset records raised an alarm before onset. The reason is the one identified in Section 4.2.2: components that start fresh see only a few months of data, over which the bias often exceeds the commissioning-based allowance, so the slack that protects the single-start process is not available to them. PHASE-R is therefore valid under the condition of Proposition 6 but not empirically calibrated by the allowance used here; to keep its error under control the allowance would have to bound the bias block by block, a larger and slower-to-estimate quantity that we have not pursued. We report PHASE-R as a tested remedy with a cost, not as a recommended method.

4.2.8 What the physics contributes: ablations and digital-twin error

Two ablations isolate roles of the physics (Table 10). PHASE-S replaces the physics-derived cone by the plain non-positive orthant (singletons, pairs and the all-modes direction in standardised residual space) with everything else unchanged, so it keeps the sign restriction and drops the finite-element sensitivities. PHASE-NA lets the surrogate adapt without withholding the identifiable damage-consistent component.

The sign restriction alone is not enough. PHASE-S kept the false-alarm control (0/300 at both horizons) but was much less powerful than PHASE: 2% of the 2% losses were detected against 34%, the 5% loss at storey 3 took 91 days against 59, the 5% loss at storey 1 57 days against 38, and after two healthy years 6% of 5% losses were detected against

Table 11: Digital-twin error. Each cell: records with a false alarm among 100 healthy records / detections of a 5% loss at storey 3 among 100 damaged records / median delay in days. The standard deviations are those of the relative errors in storey stiffness and floor mass of the monitor’s model.

Twin error	PHASE	PHASE-BB	PHASE- Ω	PHASE-NA	CTM-F	CUSUM
10% stiffness, 5% mass	0 / 100 / 60	0 / 100 / 57	0 / 21 / 122	0 / 100 / 60	0 / 100 / 72	0 / 100 / 26
20% stiffness, 10% mass	0 / 100 / 62	0 / 100 / 57	0 / 23 / 173	0 / 100 / 63	0 / 100 / 72	1 / 100 / 26

66%. In this generator the physics-derived directions, which say which combinations of frequencies a loss produces and in what proportion, therefore add power beyond the sign, and removing the sign as well (PHASE- Ω) is worse still. The caveat is that the twin is the generator’s own model; the next experiment relaxes that. PHASE-NA, by contrast, was indistinguishable from PHASE: 49 against 51 detections of the 2% loss, identical delays at 5% and 10%, and 61 against 66 detections after two healthy years. Over the horizons simulated the surrogate absorbs the confounded component too slowly (timescale about 2,000 epochs; Section 4.2.5) for withholding it to matter, and the longest post-onset window is 1,095 epochs. The benefit of identifiability-aware adaptation is therefore not demonstrated here and would need damage lasting several adaptation timescales.

Digital-twin error. In every other experiment the monitor’s twin is the generator’s own first-order model. To test twin error, the sensitivities, cone, nuisance projector and μ_0 were computed from a model whose storey stiffnesses and floor masses carried independent Gaussian errors of standard deviation 10% and 5% (and, in a second run, 20% and 10%), drawn anew for each record, while the data came from the unperturbed structure (Table 11).

None of PHASE, PHASE-BB, PHASE- Ω , PHASE-NA and CTM-F raised a false alarm at either error level (one CUSUM record did at the larger), and PHASE detected the 5% loss in all 100 records, at medians of 60 and 62 days against 59 days with the exact twin; the cone-free ablation again detected about a fifth. Twin error of this size therefore did not degrade the results. The limits are clear. The generator’s temperature law is a uniform change of modulus, so the temperature direction κ is exact in any twin (the row-sum identity holds for every model) and the errors enter through the load direction,

the cone and the projector. Non-uniform temperature effects, effects that are not stiffness changes, and a twin that lacks structural features altogether were not tested, and 100 records bound a false-alarm probability only to about 3.6%.

4.2.9 Operational and environmental translation

The guarantee in Proposition 3 converts into an operational statement of unusual clarity when its conditions hold. It is not a rate per epoch or per year but a bound over the monitoring life of one process: each structure has probability at most α of ever raising a first false alarm, however long it is monitored, so a fleet of N structures expects at most αN such alarms in total (an expectation, not a cap on the realised number). At $\alpha = 0.01$, a portfolio of 500 monitored buildings expects at most five spurious first-alarm inspection triggers over its lifetime, not per year. This is conditional on commissioning coverage and on the conditional mean bound; it concerns first alarms, and each restart after an investigation needs its own share of the budget.

Table 12 puts the monitors on a common basis. Panel A compares the probability of at least one spurious trigger in the first monitoring year, which is defined for every monitor: 1.00 for the 3σ chart, 0.007 for the horizon-calibrated chart, and for PHASE at most the budget 0.01 (observed 0 of 300). It is worth reading this plainly: at $\alpha = 0.01$ PHASE does *not* promise fewer spurious triggers in the first year than the calibrated chart; what it promises is that the budget does not grow with the horizon. Panel B gives the recurrent rate of the 3σ chart, which is the only monitor for which a recurrent rate is meaningful here. The chart exceeded its threshold at 7.6% of epochs in the undamaged records. Under an inspect-and-resume rule that counts each exceedance as a trigger and ignores further exceedances for a lockout period after an inspection, the mean number of spurious triggers per structure-year was 111 with no lockout, 91 with a one-day lockout and 37 with a one-week lockout (60 records). A Shewhart chart has no memory, so resetting it after an inspection does not change its behaviour; the rates assume that the structure stays healthy and the chart resumes as before. They are about half the figure of 180 per year that follows from dividing a year by the two-day median time to a first alarm,

Table 12: Illustrative inspection burden for a portfolio of 500 monitored structures, with placeholder mobilisation factors; it is not an estimate for any real portfolio. Panel A is on a first-alarm basis and is commensurable across monitors (PHASE: the budget $\alpha = 0.01$, a lifetime bound that is conditional on its assumptions and is not a bound for any calendar year). Panel B applies only to the 3σ chart, with the measured rate under inspect-and-resume with a one-day lockout (91 per structure-year; 37–111 for one-week and no lockout), assuming healthy structures.

	3σ chart	Calibrated chart	PHASE
<i>A. First spurious trigger</i>			
Probability of ≥ 1 in the first year	1.00	0.007	≤ 0.01 (budget)
Portfolio, expected first triggers (first year)	500	3.5	≤ 5 (lifetime budget)
tCO ₂ e at 50 kg per mobilisation	25	0.18	≤ 0.25
tCO ₂ e at 150 kg per mobilisation	75	0.53	≤ 0.75
tCO ₂ e at 400 kg per mobilisation	200	1.4	≤ 2.0
<i>B. Recurrent practice</i>			
Spurious triggers per structure-year	~ 91	—	—
Portfolio triggers per year	$\sim 45,500$	—	—
tCO ₂ e/yr at 50, 150, 400 kg	2,300; 6,800; 18,200	—	—

which overstates the rate because a median is not a mean time between triggers. The carbon translation uses a user-supplied mobilisation factor; we deliberately avoid asserting a carbon value for an inspection, since it depends on crew travel, access equipment and structure type, and asset owners should substitute locally derived factors. The arithmetic is linear.

The comparison with the calibrated chart is the honest one, and it is not primarily about carbon: both raise few first alarms, but the calibrated chart achieves this by being unable to detect damage at all in our experiments, whereas PHASE detected 5% stiffness loss at storey 3 in every record, at a median of 59 days (interquartile range 51–69), when onset followed three months of monitoring. The case for a time-uniform budget rests less on avoided inspections than on preserving the detection capability that makes early, repair-scale intervention possible, and that capability decays with the time since commissioning (Section 4.2.7). These are illustrative calculations from one simulated generator.

5 Application to the Z24 bridge

The simulation study above is controlled but synthetic. This section applies PHASE to real measurements from the Z24 progressive damage test, the canonical benchmark in vibration-based SHM (Peeters and De Roeck, 2001), in which a three-span prestressed concrete highway overpass near Koppigen, Switzerland, was subjected to a documented sequence of structural interventions before demolition in 1998; the test programme is described in detail by Reynders and De Roeck (2009).

5.1 Data and modal feature extraction

We use the released ambient-vibration records for the first ten test states: two undamaged reference measurements (4 and 9 August 1998), progressive lowering of the Koppigen pier by 20, 40, 80 and 95 mm, a foundation tilt, a third reference measurement taken after the settlement had been reversed and the pier lifted back, and concrete spalling at the soffit over 12 and 24 m². Accelerations were sampled at 100 Hz in nine instrument setups per state; we use only the five permanently installed reference accelerometers, which are common to every setup and therefore free of the roving-sensor differences between setups. Both released measurement modalities are analysed: ambient vibration, driven by traffic and wind, and forced vibration, driven by shakers. The forced records have a better signal-to-noise ratio and give within-state coefficients of variation of 0.24–0.97% against 0.35–1.4% for the ambient records, so they serve as an independent check that the conclusions are not artefacts of one excitation type.

Each record is divided into non-overlapping windows of 8,192 samples (81.9 s), each treated as one monitoring epoch. For each epoch the cross-power spectral density matrix of the five reference channels is formed by Welch’s method and its first singular value taken, that is, frequency domain decomposition; three natural frequencies are then peak-picked in fixed bands with parabolic interpolation. This yields 700 ambient and 702 forced epochs across the ten states (63 to 72 per state). The three tracked modes lie near 3.87, 9.82 and 12.7 Hz and have within-state coefficients of variation of 0.35%, 0.75% and 1.4%;

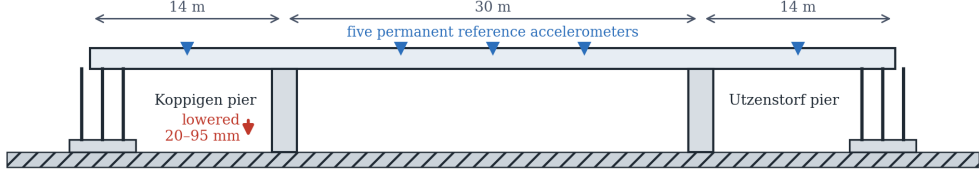


Figure 8: Schematic elevation of the Z24 bridge: a three-span prestressed concrete box-girder overpass, with the two central piers stiffly connected to the girder and column triplets at the abutments. Progressive damage was introduced by lowering the Koppigen pier in four stages, producing the settlement sequence analysed in experiment R2; the lowering was later reversed, which experiment R3 analyses. Accelerometer positions are indicative only. Drawing by the authors; dimensions follow the published description of the structure (Peeters and De Roeck, 2001; Reynders and De Roeck, 2009).

other candidate bands were discarded because the automatic picker could not track them consistently across all ten states, and this restriction to $m = 3$ features is a limitation of the analysis rather than of the data. The bands are fixed. In the 80 mm, 95 mm and tilt states 7–29% (ambient) and 8–53% (forced) of the picks for the second and third tracked modes lie within 0.05 Hz, about one frequency line (0.049 Hz), of the lower band edge, so the largest shifts may be truncated by the band, and mode identity was not checked against mode shapes.

Figure 8 shows the structure and the location of the intervention. Figure 9(left) shows the resulting state means. The physics is clearly visible: all three frequencies fall monotonically through the settlement sequence, by 4.9%, 1.3% and 4.0% respectively at 95 mm, and all three recover when the settlement is reversed.

5.2 What this dataset can and cannot test

Two features of the released data constrain what may be concluded, and we state them before reporting results. First, no environmental covariate accompanies the ambient-vibration files we used (environmental measurements exist for the Z24 monitoring year and for KW51 and would allow a test of the full surrogate; we did not use them here), so the physics-structured surrogate of Section 3.3 degenerates to a predictable running intercept and the identifiability analysis cannot be exercised; correspondingly, the damage cone is instantiated at its coarsest physically justified level, namely that stiffness loss cannot

raise a natural frequency, so the admissible set is the non-positive orthant in standardised log-frequency space. Localisation is not attempted, as no calibrated finite element model of the bridge is available to us.

Second, and more consequentially, each state was measured on a different day. The two undamaged reference measurements, taken five days apart, differ by 0.9%, 1.4% and 0.3% in the three tracked modes, which is about two to three times the within-state scatter for the first two modes and smaller than it for the third. Test-day effects are therefore of the same order as the damage signatures, and no monitor can attribute a change to damage rather than to environment without a covariate. The main results below treat the epochs within a state as exchangeable and are reported over 200 random orderings of the epochs within each block. That is an assumption, and we check it rather than assume it. For the ambient records, the median lag-one autocorrelation of the tracked log-frequencies within a state is 0.02, and only 1 of the 30 state-by-mode series is significant at the 5% level (3%, close to chance); a trend test is significant for 13% and a test for a set-up effect for 7%. For the forced records the median lag-one autocorrelation is 0.36 and 70% of the series are significant, so exchangeability within a forced state is not supported. Section 5.3 therefore also reports re-analyses that keep the recorded order (chronological, reversed), a random circular shift of the recorded order, and shuffles of whole measurement set-ups with the order inside each set-up preserved. The 200 orderings are a resampling device for this one dataset: the reported standard errors are Monte Carlo errors over orderings, a rate of 0 or 1 means the outcome does not depend on the ordering, and none of it quantifies uncertainty about other bridges or independent monitoring campaigns, of which there is one.

Which components the Z24 data validate. Table 13 lists which parts of PHASE are exercised by the simulation, by Z24, or by both.

Is the sign restriction appropriate for settlement, tilt and spalling? The cone is derived for stiffness loss at fixed mass, and the Z24 interventions are pier settlement,

Table 13: Evidence for each component of PHASE. “Reduced” means that the component is run in a simplified form because the Z24 release contains no environmental covariates and no calibrated structural model.

Component	Simulation (Section 4)	Z24 (this section)
Physics-structured surrogate with tied temperature response	Yes; compared with the black-box surrogate	No: reduced to a running intercept
Damage cone (sign restriction)	Yes; ablations PHASE- Ω (no sign) and PHASE-S (sign only), benign stiffening	Reduced: non-positive orthant; directional discrimination on the recovery state
Nuisance subspace, identifiable fraction, localisation	Yes; localisation correct at storeys 3 and 6, failed at storey 1	No: cannot be exercised; localisation not attempted
Identifiability-aware adaptation	Used in every run; ablation PHASE-NA shows no measurable effect over the horizons simulated	Reduced: filtered intercept update with the orthant cone
Betting e-process with allowance $\hat{\delta}$	Yes; no-allowance and certified variants	Yes, with $\hat{\delta}$ from the commissioning data
Commissioning window and envelope	Yes; sweep of window length	Yes; depends on how commissioning data are ordered
Separation of damage from environment	Yes, for a generator with unmodelled effects	No; the data cannot test it

foundation tilt and concrete spalling, which alter support conditions, load paths and cross-section as well as stiffness. The sign restriction is a modelling assumption, and for these interventions it is supported by the data only a posteriori. Relative to the pooled reference states, the three tracked frequencies fell in every damage state of both modalities from 80 mm settlement onwards: by up to 5.5%, 2.3% and 4.0% at 95 mm, by 1.0–1.3%, 1.2–1.9% and 3.4–3.7% under foundation tilt, and by 0.5–0.8%, 0.8–1.2% and 1.5–2.1% under spalling. These spalling figures are relative to the pooled reference states, but state 8 (reference 3) already lies 0.9–1.3% below them; relative to state 8 the first two tracked frequencies change by between -0.1% and $+0.4\%$ under spalling, and only the third falls, by 0.6–1.6%. Only the third mode in the two smallest settlement states rose, by at most 0.6%, comparable to the day-to-day offset. That agrees with the physical expectation that settlement and tilt release restraint and that spalling removes section stiffness faster than mass, but it does not justify the assumption in general. Mechanisms that can *raise* some tracked frequencies, such as mass loss, crack closure under prestress or tempera-

Table 14: Z24 bridge: alarm rate over 200 random epoch orderings; median alarm epoch in parentheses for R2, where timing distinguishes the methods (all methods shown; CUSUM, whose threshold is calibrated on the commissioning window alone, alarms early and also alarms in 86% of R1b orderings). Nominal level $\alpha = 0.01$; Monte Carlo standard errors over orderings, at most 0.035 (these describe the resampling of one dataset, not independent campaigns). PHASE- Ω is the cone-free ablation, CTM the conformal test martingale, MSD the Mahalanobis novelty index. EWMA and Shiryaev–Roberts, omitted for space, behave like CUSUM. Without replicate healthy campaigns, all thresholds here are calibrated on the commissioning window alone, as a practitioner would have to do.

Commissioning $\rightarrow$ monitored state	PHASE	PHASE- Ω	CTM	CUSUM	3σ	χ^2	PCA	MSD
<i>ambient vibration (700 epochs)</i>								
R1a ref 1 $\rightarrow$ ref 2 (new day)	1.00	1.00	1.00	1.00	1.00	1.00	0.97	0.90
R1b ref 1+2 $\rightarrow$ held-out ref.	0.00	0.00	0.01	0.86	0.07	0.14	0.00	0.68
R2 ref 1+2 $\rightarrow$ settlement 20–95 mm	1.00 (72)	1.00 (119)	1.00 (76)	1.00 (12)	1.00 (78)	1.00 (75)	1.00 (105)	1.00 (143)
R3 95 mm $\rightarrow$ reversed (recovery)	0.00	1.00	0.00	0.00	1.00	1.00	1.00	1.00
R4 ref 1+2 $\rightarrow$ spalling 12, 24 m ²	1.00	1.00	1.00	1.00	1.00	1.00	0.53	0.22
R5 ref 1+2 $\rightarrow$ reference 3 (new day)	1.00	1.00	1.00	1.00	1.00	1.00	0.52	0.18
<i>forced vibration (702 epochs)</i>								
R1a ref 1 $\rightarrow$ ref 2 (new day)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
R1b ref 1+2 $\rightarrow$ held-out ref.	0.00	0.00	0.01	0.86	0.42	0.35	0.00	0.00
R2 ref 1+2 $\rightarrow$ settlement 20–95 mm	1.00 (29)	1.00 (103)	1.00 (31)	1.00 (5)	1.00 (86)	1.00 (50)	1.00 (144)	1.00 (144)
R3 95 mm $\rightarrow$ reversed (recovery)	0.00	1.00	0.00	0.00	1.00	1.00	1.00	1.00
R4 ref 1+2 $\rightarrow$ spalling 12, 24 m ²	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.87
R5 ref 1+2 $\rightarrow$ reference 3 (new day)	1.00	1.00	1.00	1.00	0.84	0.95	0.91	0.14

ture, or boundary stiffening, lie outside the cone, and a net frequency increase caused by damage would not be flagged as damage. As implemented, the score is the maximum over singleton, pair and all-mode generators, so a pattern in which some frequencies fall and others rise still scores on the falling ones; it is only the rising components that are discarded. Two mitigations follow. The cone can be enlarged with generators for known mass-change mechanisms, at the cost of power. And an omnibus process (PHASE- Ω here) can run alongside as a separate *change notification* channel with its own share of the budget, so that a frequency rise prompts engineering review without being reported as damage.

5.3 Results

Table 14 reports six experiments.

The commissioning envelope on real data. R1a commissions the monitor on the first reference measurement and then monitors the second, five days later, with no damage in between. Every method, PHASE included, alarms in every ordering under both

excitation types. The estimated bias discount, $\hat{\delta} = 0.13\text{--}0.14$, was calibrated within a single test day and does not cover between-day variation. R1b repeats the exercise with the commissioning window pooled across both reference days, with epochs from the two days mixed at random, and monitoring held-out reference epochs: PHASE, its omnibus variant and the PCA chart then alarm in none of 200 orderings, the conformal martingale in 1%, while the repeated χ^2 test alarms in 14% of ambient and 35% of forced orderings, and CUSUM and Shiryaev–Roberts – whose thresholds a practitioner without replicate healthy campaigns must calibrate on the commissioning window – alarm in 86% and 83%. The simulation finding of Section 4.2.6, that monitoring with a commissioning-based allowance is invalid when commissioning does not cover the conditions it will meet, is thus consistent with the real measurements in both modalities. R1b is a single random split of the same two days (84 epochs per half drawn at random from the pooled reference days), so it shows that a window mixing both days covers held-out epochs of those same days, not that it covers a new day; R5 is the test of a new day.

Directional behaviour on real data. R3 is the real-data counterpart of the benign stiffening experiment. Commissioning on the 95 mm settled state and then monitoring the state after the settlement had been reversed, PHASE alarmed in none of 200 orderings while the otherwise identical omnibus variant alarmed in all of them, at a median of 14 epochs. The 3σ chart, the repeated χ^2 test, the PCA chart and the Mahalanobis index also alarmed in every ordering; the result is identical under forced excitation. The bridge really did change, and substantially: all three frequencies rose relative to the 95 mm settled state, by 4.6%, 0.9% and 2.7% (they remain 0.9–1.3% below the original reference states in the first two modes, so “recovery” is partial). A damage detector should not report damage when a structure recovers stiffness, and the physics-derived sign restriction is what supplies that discrimination. That silence is guaranteed by construction for any monitor that scores only frequency decreases, so R3 does not test the skill of the detector; it shows that the real data contain a substantial benign change which omnibus detectors report as damage, the practical consequence of ignoring the sign. The conformal martingale,

CUSUM, Shiryaev–Roberts and EWMA also stay silent here, but they consume the same cone score, so their immunity is inherited from the physics rather than earned by their own machinery. Because the restriction is directional, this conclusion is immune to the attribution difficulty discussed below: whatever caused the frequencies to rise, damage did not. It is also the one result that is insensitive to the ordering of the data: PHASE stayed silent and PHASE- Ω alarmed in every ordering of Table 15, in both modalities.

Detection depends on how commissioning data are ordered. In R2 the monitor is commissioned on both reference days and then sees the settlement sequence. With the epochs of the two days mixed at random, PHASE alarms in every ordering, at a median of 72 epochs under ambient excitation and 29 under forced excitation, where the lower identification scatter roughly halves the delay; the omnibus variant alarms later, at 119 and 103 epochs, and the data-normalisation charts later still. Re-analysis with dependence preserved (Table 15) shows what this result does and does not depend on. Shuffling whole measurement set-ups across the two reference days, with the order inside each set-up kept, reproduces the main table (PHASE at 58 and 30 epochs, the omnibus variant at 120 and 105), so serial dependence within set-ups is not what drives it. Presenting the commissioning data in the order in which they were recorded, the first reference day followed by the second, changes the picture completely: the first half of the commissioning window then fixes the intercept and the second half, a different day, fixes the scale and the allowance, which rises to 0.61 (ambient) and 0.58 (forced) from mean values of 0.06 over the mixed orderings, nearly the whole range of the score above μ_0 . PHASE then does not alarm at all on the ambient settlement sequence and alarms at epoch 189 on the forced one, later than the omnibus variant (127). The same holds for the other orderings that keep the two days apart (reversed order, circular shifts, shuffled set-ups within each day). Table 14 also shows that the advantage of PHASE under mixed commissioning is over the omnibus variant and the data-normalisation charts only: the memoryless 3σ and χ^2 charts alarm at about the same epoch as PHASE (78 and 75 against 72 for ambient excitation), and CUSUM and Shiryaev–Roberts, whose thresholds

Table 15: Z24: sensitivity of PHASE and PHASE- Ω to the ordering of the data. Each cell gives PHASE / PHASE- Ω as alarm rate over 200 orderings (median alarm epoch). Random: uniformly random order within and across the commissioning states (main table). Set-ups: whole measurement set-ups shuffled across the commissioning days, order inside a set-up kept. Recorded: the order in which the data were measured (the first reference day, then the second). Shift: random circular shift of the recorded order within each state. For the recorded order only the conformal martingale involves randomness.

	Random	Set-ups	Recorded	Shift
<i>ambient vibration</i>				
R1a	1.00 (10) / 1.00 (15)	1.00 (11) / 1.00 (16)	1.00 (10) / 1.00 (14)	1.00 (10) / 1.00 (16)
R2	1.00 (72) / 1.00 (119)	1.00 (58) / 0.98 (120)	0.00 / 1.00 (197)	0.00 / 1.00 (197)
R3	0.00 / 1.00 (14)	0.00 / 1.00 (14)	0.00 / 1.00 (14)	0.00 / 1.00 (12)
R4	1.00 (11) / 1.00 (25)	1.00 (12) / 0.94 (27)	0.00 / 0.00	0.00 / 0.04 (140)
R5	1.00 (11) / 1.00 (25)	1.00 (11) / 0.96 (25)	0.00 / 0.00	0.00 / 0.00
<i>forced vibration</i>				
R1a	1.00 (10) / 1.00 (13)	1.00 (12) / 1.00 (14)	1.00 (13) / 1.00 (12)	1.00 (11) / 1.00 (12)
R2	1.00 (29) / 1.00 (103)	1.00 (30) / 0.96 (105)	1.00 (189) / 1.00 (127)	1.00 (189) / 1.00 (130)
R3	0.00 / 1.00 (10)	0.00 / 1.00 (10)	0.00 / 1.00 (9)	0.00 / 1.00 (10)
R4	1.00 (11) / 1.00 (19)	1.00 (12) / 0.95 (20)	0.00 / 1.00 (83)	0.00 / 1.00 (83)
R5	1.00 (11) / 1.00 (23)	1.00 (12) / 0.95 (24)	0.00 / 1.00 (54)	0.00 / 1.00 (54)

are calibrated on the commissioning window alone and which false-alarm in R1b, alarm much earlier. The finding that the cone-restricted process responds sooner to real damage than the omnibus process is therefore supported only when commissioning epochs from the two days are mixed, which is an exchangeability assumption across test days that a real deployment would not satisfy; we do not claim it as a general finding.

Attribution. R5 forbids reading any R2 alarm as damage detection. Monitoring the third reference measurement, taken on yet another day after the settlement was reversed, produces an alarm in every ordering for PHASE when commissioning epochs are mixed (median epoch 11), but in none when the commissioning days are kept apart, where the inflated allowance absorbs the between-day difference; the omnibus variant alarms under forced excitation (epoch 54) but not under ambient. That state is nominally undamaged, though it is documented as not identical to the original condition, the pier support having been altered by the lifting operation. No ordering gives a monitor that is silent on this undamaged day and also detects the settlement sequence promptly. Under mixed commissioning, R4 (spalling) and R5 (a reference day) also give identical outcomes, an alarm

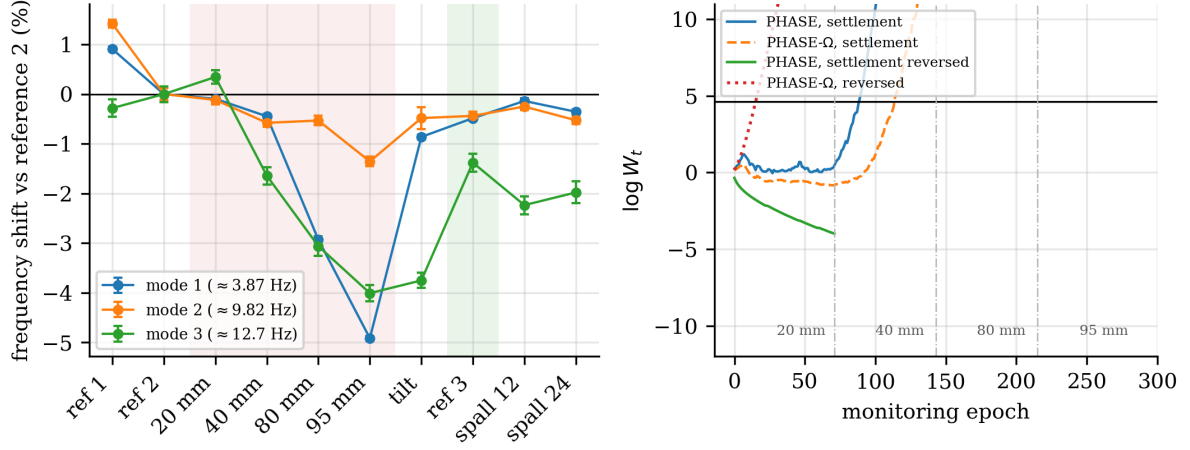


Figure 9: Z24 bridge. Left: mean shift of the three tracked natural frequencies in each structural state relative to the second reference measurement, with standard errors; shading marks the settlement sequence (red) and the state after the settlement was reversed (green). Right: accumulated log-evidence for PHASE and the omnibus variant under the settlement sequence and under the reversal, with the alarm threshold.

in every ordering at median epoch 11, so the spalling alarm cannot be distinguished from a new-day effect. We therefore report the R2 and R4 results as showing that PHASE responds to real structural change under mixed-day commissioning, and explicitly not as evidence that it separates damage from environment on this dataset. That separation requires the covariate stream that the Z24 release does not include.

5.4 What the real-data study adds

Two claims of the paper survive contact with real measurements, and a third does not. The dependence of the monitor on its commissioning data is confirmed, as a failure mode of PHASE and not only of its competitors: a monitor commissioned within one test day alarms on the next, and the allowance estimated from commissioning data changes about ten-fold depending on whether the commissioning window contains one day or two kept apart. The value of restricting the alternative to physically admissible directions is confirmed in the cleanest possible form, on a genuine structural change of known sign, under every ordering and both excitation types. The claim that the cone-restricted process crosses the alarm threshold earlier than the omnibus process on a real damage sequence holds only when commissioning epochs from the two reference days are mixed, and is not

claimed as a general finding. What the dataset cannot support, and what we therefore do not claim, is calibrated separation of damage from environmental variation: that would require a monitoring campaign with a synchronous covariate record, of which the Z24 long-term monitoring series and the KW51 campaign are the obvious candidates for future work. Of the components of PHASE, only the cone, the e-process and the allowance are exercised here, in reduced form (Table 13); the physics-structured surrogate, the nuisance subspace, identifiability-aware adaptation in its full form and localisation are validated by simulation only.

6 Discussion and conclusions

6.1 Implications for practice

Three practical implications follow, each conditional on the assumptions stated.

First, alarm governance can become a budget rather than a tuning exercise, *if* commissioning covers the operating envelope and the conditional-mean bound holds. Under those conditions an asset owner can state that each structure has probability at most 1% of ever raising a first false alarm, so that a fleet of N structures expects at most $0.01N$ of them. That is a candidate specification for a performance agreement, but the present evidence does not support it as an unconditional contractual commitment. The calibration procedure has not been shown to enforce the conditions (Proposition 4; Section 4.2.6); the variant with a provable calibration step (PHASE-C) pays a large price in power; and the guarantee covers the first alarm of one monitoring process, so an agreement would have to define the commissioning protocol, a coverage test, the restart rule and how the budget is spent across restarts (for example by giving the k -th restart a share α_k with $\sum_k \alpha_k \leq \alpha$). Because the guarantee is uniform over stopping rules, operators may watch the evidence process continuously without invalidating the alarm. That does not extend the guarantee to action rules set below the alarm threshold: by Ville’s inequality the probability that the evidence ever reaches a level w is at most $1/w$ under the null, so an

inspection policy triggered at $W_t \geq 10$ is protected only at the level 0.1, not 0.01. The budget concerns false alarms only: silence is not evidence that a structure is undamaged, because power depends on the severity of the damage, on the delay and on the time since commissioning, and missed detections are not controlled by α . The choice of α trades the cost of a false inspection against the cost of delay: by Proposition 5 the delay bound grows with $\log(1/\alpha) + \log |\Lambda|$, which is 7.1 at $\alpha = 0.01$ with twelve bets, 5.5 at $\alpha = 0.05$ (a bound about 23% shorter) and 9.4 at $\alpha = 0.001$ (about 32% longer); α should be set from those costs and not by convention.

Second, the detectability map belongs in the design phase. It is computed from the finite element model and the expected identification scatter, before installation, and states which substructures a proposed sensor and mode configuration can support claims about *relative to the nuisance model assumed*. In our eight-storey frame, moving from four to six identified modes raised the identifiable fraction at storeys 1 and 8 from 0.034 and 0.064 to 0.245 and 0.349, converting effectively blind substructures into monitorable ones. The map is not a localisation guarantee: storey 1 has a non-zero identifiable fraction and was nevertheless never localised correctly (Section 4.2.5), because localisation also needs the identifiable signatures of different storeys to be separable, and the map changes when the nuisance model changes (Table 7). This is a first-pass, physics-derived criterion for instrumentation planning, and it integrates naturally with BIM-based asset models where the finite element model and sensor layout already coexist.

Third, the commissioning requirement should be treated as a design constraint rather than a nuisance. Our results indicate that a monitoring system commissioned over a month or two was invalid in our sweep, and over a few months valid but nearly powerless. What is required is adequate coverage of the operating envelope; in the seasonal scenario simulated here a full calendar year was what achieved that, which is not a universal requirement. Section 6.2 sets out how coverage can be assessed in practice. Where waiting a full year is unacceptable, the natural remedy is exactly the one this special issue is concerned with: use the calibrated digital twin to *synthesise* the operational envelope, generating reference-condition modal responses across temperatures and load

states that have not yet been observed, and calibrate $\hat{\delta}$ against simulated rather than measured excursions. This substitutes model risk for extrapolation risk, and quantifying that trade-off is a natural next step.

6.2 Deployment prerequisites and practical considerations

Prerequisites. PHASE needs a digital twin with modal sensitivities, at least $\dim \mathcal{N} + 1$ reliably identified modes (more for localisation), synchronous measurements of the covariates the nuisance model uses, an automated modal identification chain with quality control, and a commissioning period that is declared complete only when the coverage checks below pass.

Judging commissioning coverage in advance. Representativeness refers to the future, so it cannot be certified from commissioning data alone; the practical question is how to make the assumption checkable. We suggest, and have only partly tested, the following protocol. (i) Declare the target envelope before commissioning from independent records, for example multi-year site climatology for temperature and the occupancy or traffic schedule for load, and measure the fraction of that envelope covered by the commissioning data. In the simulated sweep (Table 8) validity returned only when coverage of the monitoring epochs was close to complete; at coverage around one half, the false-alarm probability was 0.24–0.25. (ii) Estimate $\hat{\delta}$ separately on seasonal or operational blocks of commissioning data and treat a large spread as evidence that one block does not represent the others. (iii) Report coverage online and flag, or gate, alarms issued outside the commissioning envelope. Gating by itself did not restore validity in our sweep (PHASE-E), so this is a diagnostic, not a remedy. (iv) Treat the guarantee as inactive, and alarms as provisional, until declared coverage is reached; extend commissioning, as a restart with its own budget share, when a season or operating regime not yet seen occurs. (v) Where the envelope cannot be observed, use the twin to synthesise it and state the model risk. None of these steps converts representativeness into a theorem; they make a failure visible earlier.

Missing data and recalibration. Skipping an epoch leaves the evidence process unchanged provided the decision to skip is made without looking at that epoch’s score (sensor outage, a scheduled quality rejection based on the identification’s own uncertainty); screening epochs by the size of their residual would bias the process and void the guarantee. Dropped or mismatched modes were not simulated. Any re-estimation of the scatter D , of μ_0 or of $\hat{\delta}$, and any re-commissioning after maintenance or a retrofit, starts a new process and should consume a pre-allocated share of the budget; known changes outside the cone, such as stiffening retrofits, should trigger planned re-commissioning.

Long-lived assets. Power decays with the healthy time since the last start of the evidence process (Section 4.2.7), so a single commissioning followed by decades of monitoring is not a configuration we can recommend. A natural configuration, which we have not evaluated, is scheduled re-commissioning, for example annually, in which the k -th restart spends a share $\alpha_k = 6\alpha/(\pi^2 k^2)$ of the lifetime budget (so that $\sum_k \alpha_k = \alpha$) at the price of a higher threshold $1/\alpha_k$ for later restarts. The restart variant PHASE-R, which achieves a similar effect without re-commissioning, lost its empirical error control. Persistent benign changes, such as a retrofit that stiffens the structure, should trigger re-commissioning; the effect of a long silent benign period on later detection was not simulated.

Are the delays acceptable? A median delay of 59 days for a 5

6.3 Sustainability implications

The sustainability case for instrumented infrastructure rests on the claim that continuous monitoring enables condition-based intervention: repairs made early, at small scale, in place of replacement. That claim depends on alarms being both trusted and timely. Our results indicate that the conventional charts implemented here can satisfy neither simultaneously – naive charts alarm constantly, and charts calibrated to stop alarming stop detecting – while an anytime-valid rule can, in the simulated setting and conditionally

on adequate commissioning. The relevant environmental benefit would therefore not be primarily the avoided carbon of unnecessary inspections, real though that is, but the preservation of the embodied carbon in structures that are repaired rather than replaced because deterioration was detected while it was still small. That benefit is a hypothesis motivated by these results; this paper measures false-alarm and detection behaviour, not avoided replacements or avoided emissions. Reviews of AI in the built environment tend to locate sustainability gains in energy optimisation and material efficiency (Bibri and Huang, 2025); the gain argued here is of a different kind, residing in the trustworthiness of the decisions a monitoring system supports. This connects to SDG 9 through resilient infrastructure, SDG 11 through the safe operation of the existing building stock, and SDG 13 through the emissions avoided by extending service life.

There is a further, less obvious possible implication for equity. The commissioning result suggests that rigorous monitoring requires patience rather than only expensive hardware, and the physics-informed surrogate delivers usable behaviour from many fewer parameters than a black-box alternative during short commissioning, although with a long commissioning window the black-box surrogate matched it. Both features could favour deployment in rapidly urbanising contexts where instrumentation budgets are constrained but the building stock is young and its deterioration curve has not yet been characterised. These are plausible motivations, not demonstrated outcomes: the study contains no cost data, and a year of commissioning with covariate sensing and a calibrated twin is not free.

6.4 Limitations

The controlled evidence here is simulation-based and the real-data study is limited in the ways set out in Section 5.2: three tracked modes, no covariate record, no localisation, and states measured on different days so that detection cannot be attributed to damage rather than environment. The generator was constructed to be adversarial in specific ways – unmodelled thermal lag and gradients, heavy-tailed identification noise, an imperfect load proxy – but a shear frame is a simple structure, real EOVS includes effects we did not model such as humidity, wind and freeze–thaw stiffening, and real modal identification

produces mode-mismatching and dropouts that we did not simulate. The Z24 progressive damage test exercises the directional claim on real measurements and the commissioning claim only in part, and it cannot exercise the identifiability analysis; a campaign with synchronous temperature, such as the Z24 long-term monitoring series or the KW51 retrofitting campaign, is the necessary next step. The monitor’s twin is the generator’s own first-order model; an experiment with perturbed stiffness and mass (Section 4.2.8) did not degrade the results, but the generator’s temperature law is a uniform change of modulus, so non-uniform temperature effects and a structurally incomplete twin remain untested, and real EOV is richer than simulated. The comparators are single implementations calibrated on simulated healthy campaigns; their behaviour here should not be generalised to environmental-normalisation methods as a class.

Several limitations are methodological. Only frequencies are used; mode shapes and their derivatives carry localisation information that the cone construction could exploit directly. The null (10) is stated as a conditional mean bound that the surrogate is meant to make true, not a property the physics guarantees, and the weight falls on $\hat{\delta}$. The theorem holds conditionally; the implemented estimate of $\hat{\delta}$, two standard errors from a finite commissioning sample, has no guarantee of its own, and the provably calibrated variant PHASE-C is too conservative to be useful over a one-year window. The representativeness assumption that would close the gap cannot be verified from commissioning data. Our commissioning sweep shows how the allowance fails when the envelope is not covered. The long healthy period costs power: after two years PHASE detected 5% loss in two thirds of records and 2% loss in none, and the restart variant that repairs this lost its empirical false-alarm control. The detection delays reported are two to three times those of an oracle CUSUM with early onset and, after two healthy years, about eight times (203 against 27 days for 5% loss), which is the price of the anytime guarantee and of the conservative allowance, and for a structure monitored under a genuinely fixed horizon by an analyst who never peeks, the oracle procedure would be preferable. Finally, the localisation index is a heuristic accumulator rather than a calibrated estimator, with a baseline preference for some storeys on healthy data; giving localisation claims their own

anytime-valid guarantee via e-value based false discovery rate control across zones is a clear extension.

6.5 Conclusions

Structural health monitoring systems make thousands of decisions per year using rules designed for one. This paper has shown that, for the charts implemented here, the resulting false-alarm probability is not a small correction – a 3σ chart alarmed on every undamaged simulated structure within days – and that raising thresholds to control error over one monitoring year traded the problem for a worse one, since the chart so calibrated detected no 5% damage at all in our experiments.

PHASE addresses this by replacing the threshold with a betting e-process whose guarantee holds at every instant of an unbounded campaign, conditionally on a bound for the expected score of an undamaged structure, and by using eigenvalue perturbation in four distinct roles: constraining the nuisance surrogate, restricting the alternative to a polyhedral damage cone, determining which stiffness-loss patterns are identifiable relative to the nuisance, and protecting identifiable damage from online adaptation. The evidence for the four roles differs: the physics-derived cone added power beyond a plain sign restriction (and that comparison survives errors of 10–20% in the twin), the physics-structured surrogate helped only when commissioning was short, the detectability map predicted neither detection nor localisation reliably, and withholding identifiable damage from adaptation showed no measurable effect over the horizons simulated. Across 4,000 simulated records it raised no false alarm on 300 undamaged structures at either a one- or a three-year horizon (upper 95% limit 1.2% in each), detected 5% single-storey stiffness loss at storey 3 in every record within a median of 59 days (and 2% loss in a third of records), localised damage correctly at storeys 3 and 6 but not at storey 1, and remained silent under a benign stiffening event that an otherwise identical omnibus detector flagged in every record. The observed false-alarm control belongs to the implemented, empirically calibrated rule; the theorem covers the conditional statement, and the provably calibrated variant has little power. Damage that starts after a long healthy period is detected more slowly and

less often, and a restart variant that removes most of that penalty exceeded the nominal false-alarm level.

On real measurements from the Z24 bridge the data support two narrow conclusions: a monitor commissioned within a single test day false-alarms on the next undamaged day, and when the pier settlement was reversed and the bridge recovered stiffness, PHASE stayed silent in all 200 orderings while every omnibus detector reported damage in all of them. Its response to settlement depends on how commissioning epochs are ordered and is not evidence of separation between damage and environment.

Two findings qualify the method and, we believe, matter more than the headline comparisons. Stiffness-loss patterns whose signature lies in the nuisance subspace – uniform loss always, and for the frame studied here a four-dimensional family including equal loss at storeys 1 and 8 – cannot be distinguished from temperature or load change using frequencies, so a monitor should state which substructures it can and cannot speak about, and identifiability relative to the nuisance is not the same as localisability. And calibrated monitoring requires a commissioning window spanning the operational envelope; with 30 to 60 days of commissioning the monitors tested here were invalid, and up to about 300 days the commissioning-based allowance left them valid but nearly powerless. Both limitations are consequences of the physics, and both are visible only because the physics was carried into the inference rather than left in the surrogate; the second is confirmed on Z24 and the first only in simulation. The Z24 study adds a third: without a synchronous environmental covariate, no monitor can attribute a change to damage, and reporting detection rates on such data risks crediting the detector for the calendar.

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A Notation and acronyms

Structure and physics

n, m	number of substructures (storeys) and of identified modes
K_j, M, θ_j, ρ_j	substructure stiffness matrix, mass matrix, stiffness multiplier, stiffness loss $\rho_j = 1 - \theta_j \geq 0$
$\lambda_i, \omega_i, \phi_i$	eigenvalue, circular frequency and mode shape of mode i
S, κ, η	sensitivity matrix of log-frequencies to stiffness loss (eq. 2); response to uniform modulus change ($\kappa = S\mathbf{1}$); response to added live-load mass
$D, A = D^{-1}S$	diagonal matrix of identification scatter; standardised sensitivity matrix
$\mathcal{C}, \tilde{d}_j$	damage cone (eq. 5) and its generators
$\mathcal{N}, P_\perp, \mathcal{K}$	nuisance sensitivity subspace, projector onto its orthogonal complement, subspace of confounded loss patterns
ϱ_j	identifiable fraction of substructure j (eq. 6)

Monitor

y_t, z_t, r_t	log-frequencies, covariates (temperature deviation ΔT_t , load proxy q_t), standardised plug-in residual
g_β, b_T, b_q	surrogate (eq. 7) with temperature coefficient b_T and load coefficient b_q
V, c, X_t	polyhedral generators of the cone; score-saturation scale c ; bounded score in $[0, 1]$ (eq. 9)
ς, ν	Huber clipping scale and forgetting factor of the adaptation
μ_0	null mean of the score for ideal standard-normal residuals
m_t	conditional mean excess $\mathbb{E}[X_t \mathcal{F}_{t-1}] - \mu_0$
$\delta, \hat{\delta}, \hat{\delta}_\varepsilon$	bias allowance; commissioning estimate (two standard errors); certified allowance (Proposition 4)
Λ, λ	grid of bets, one bet
$W_t^\lambda, W_t, \tilde{W}_t$	wealth of one bet, mixture wealth, discounted mixture wealth
$\alpha, \varepsilon, \tau$	false-alarm budget, calibration-failure budget, alarm time
σ	first epoch at which the allowance is too small ($m_t > \hat{\delta}$; Proposition 4); not the identification scatter σ_j in D
$C, n, \bar{X}_C, \bar{m}_C$	scoring epochs of the second half of commissioning, their number, the average score and the average conditional excess over them
$\tilde{W}_t^\lambda, L_t^\lambda$	discounted wealth of bet λ and its logarithm
$\gamma_\lambda, B_\lambda, b$	guaranteed log-growth rate after the change, upper bound on log-increments, and $\log(1/\alpha) + \log \Lambda $ (Proposition 5)
s_k, π_k, W_t^R	start times, weights and mixture wealth of the restart variant (Proposition 6)
t_0, t^*	end of commissioning, damage onset

Acronyms and variant labels

SHM, EOVS	structural health monitoring; environmental and operational variability
SAVI	safe anytime-valid inference
FWFA	family-wise false-alarm probability (eq. 1)
PHASE	full method; -ND no allowance; -C certified allowance; -R restart mixture; -Ω omnibus score (no cone); -S sign-only score (plain negative or-thant instead of the physics-derived cone); -NA adaptation without withholding the identifiable damage component; -BB black-box surrogate; -E envelope-gated
CTM, CTM-F	conformal test martingale with second-half or full-window calibration set
CUSUM, SR, EWMA	cumulative-sum, Shiryaev–Roberts and exponentially weighted moving-average charts
PCA-EOV, MSD	principal-component environmental-removal chart; Mahalanobis novelty index

B Implementation details

All code is supplied with the submission (`code/`); every stage uses fixed seeds. *Generator*: eight-storey shear frame, storey mass 2.5×10^5 kg, storey stiffness 6×10^8 N/m; $m = 6$ modes; identification noise Student- t_5 scaled to unit variance, relative standard deviations 0.60, 0.72, 0.84, 0.96, 1.08, 1.20% for modes 1–6; $\beta_T = 2.5 \times 10^{-3} \text{ }^\circ\text{C}^{-1}$; ambient temperature $24 - 7 \cos(2\pi(t/\text{year} - 0.05))$, diurnal amplitude 4°C , AR(0.6) weather noise with innovation s.d. 1.1°C ; storey thermal lags 0.7 to 2.0 epochs and gradient factors 0.86 to 1.16 (linear in storey); live load on floors 2–7, relative added mass $0.045 \exp(-\frac{1}{2}((h-12)/6)^2)$ times a weekday factor (1; 0.25 at weekends) plus noise 0.004, observed through a proxy with 10% multiplicative and 0.0015 additive error; epoch length 6 h (1,460 epochs per year).

PHASE. $t_0 = 1,460$ epochs. Surrogate fitted by recursive least squares with ridge 10^{-6} and forgetting factor $\nu = 0.9995$. Commissioning pass A (first 730 epochs): unfiltered fitting; the scatter D is estimated robustly from the last two thirds of the pass-A one-step residuals (median absolute deviation, residuals beyond six robust standard deviations trimmed). Pass B (last 730 epochs): the monitoring update rule runs and the scores used for $\hat{\delta}$ are sequential plug-in scores. V : 92 unit vectors (8 generators and 28 pairs with weights 0.25, 0.5, 0.75). Score scale $c = 3$, $\mu_0 = 0.3027$ (Monte Carlo, 4×10^5 draws, standard error 4×10^{-4}). Bet grid Λ : 12 values geometrically spaced from 0.05 to $0.9/\mu_0 = 2.97$, equal weights. Huber clip $\varsigma = 3$ applied after withholding the identifiable damage-consistent component. Allowance $\hat{\delta} = (\bar{X} - \mu_0 + 2\hat{s})^+$, $\hat{s} = s/\sqrt{n}$ with $n = 730$. Alarm at $W_t \geq 1/\alpha$, $\alpha = 0.01$. *Variants*. PHASE-ND: $\hat{\delta} = 0$. PHASE-C: $\hat{\delta}_\varepsilon$ with $\varepsilon = 0.005$ and threshold 200. PHASE- Ω : score $\min(\|r_t\|/8, 1)$, its own $\mu_0 = 0.2936$ and allowance. PHASE-BB: per-mode quadratic regression on $[1, \Delta T, \Delta T^2, q, \Delta T q]$, same forgetting, own scale and allowance. PHASE-E: epochs with ΔT outside the 0.5–99.5% commissioning quantiles or q above its 99.5% quantile are skipped (no bet, no discount).

Comparators. 3σ : $\max_i |r_{it}| > 3$. Calibrated chart, CUSUM, Shiryaev–Roberts, EWMA, PCA-EOV and Mahalanobis: threshold equal to the 99th percentile of the monitoring-period maximum of the statistic over the 150 calibration records (one-year horizon; calibrated chart threshold 24.32, CUSUM 17.92). CUSUM: $S_t = \max(0, S_{t-1} + X_t - \mu_0 - 0.05)$. Shiryaev–Roberts: $R_t = (1 + R_{t-1}) \exp(4(X_t - \mu_0 - 0.05))$, statistic $\log(1 + R_t)$. EWMA: smoothing weight 0.05 on X_t , standardised by its asymptotic standard deviation. PCA-EOV: two leading principal components of the commissioning log-frequencies removed, statistic the norm of the remainder divided by its commissioning standard deviation. Mahalanobis: $\sqrt{r^\top \hat{\Sigma}^{-1} r}$, $\hat{\Sigma}$ from pass-A standardised residuals. Repeated T^2 : $r_t^\top r_t > \chi_{6,0.99}^2$. CTM and CTM-F: as in Section 4.1.3, calibration sets of 730 and about 1,200 scores; ties randomised with fixed seeds.

Z24. Three tracked modes (bands 3.60–4.10, 9.60–10.15, 12.05–12.95 Hz), $m = 3$, 81.9 s epochs. Surrogate reduced to a running intercept with forgetting 0.995; cone generators: 3 singletons, 9 pairs and the all-modes direction (13 vectors), the non-positive orthant; $c = 2.5$, $\mu_0 = 0.380$; allowance and bet grid as for the simulation; comparator thresholds from the commissioning window only (99th-percentile rules scaled by 1.5 for PCA, Mahalanobis and EWMA; fixed 6.0 for Shiryaev–Roberts). Commissioning epochs are split into halves: the first half fixes the intercept and the second half the scale and $\hat{\delta}$.

C Proofs

Proposition 1. By (3) the standardised signature of a loss pattern ρ is $-A\rho$. A uniform modulus change of relative size ϵ produces $-\frac{1}{2}\epsilon D^{-1}\kappa$ and a relative added mass ζ produces $-\frac{1}{2}\zeta D^{-1}\eta$, so the set of signatures attainable by the nuisance is exactly $\mathcal{N}$. Hence (a) ρ is confounded if and only if $A\rho \in \mathcal{N}$, that is $P_\perp A\rho = 0$; the confounded patterns form $\mathcal{K} = \ker(P_\perp A)$, of dimension $n - \text{rank}(P_\perp A) \geq n - \text{rank}(P_\perp) = n - (m - \dim \mathcal{N})$. (b) For $\rho = \rho_0 \mathbf{1}$, (4) gives $S\mathbf{1} = \mathbf{1} = \kappa$, so $A\rho = \rho_0 D^{-1}\kappa \in \mathcal{N}$ for every ρ_0 . (c) Suppose A is injective and $\mathcal{N} \cap \text{range}(A) = \text{span}\{D^{-1}\kappa\}$. If $\rho \in \mathcal{K}$ then $A\rho \in \mathcal{N} \cap \text{range}(A)$, so $A\rho = cA\mathbf{1}$ (because $A\mathbf{1} = D^{-1}\kappa$), and injectivity gives $\rho = c\mathbf{1}$. Conversely, if $v \in \ker A$, $v \neq 0$, then $Av = 0 \in \mathcal{N}$ so $v \in \mathcal{K}$, and $v \not\propto \mathbf{1}$ because $A\mathbf{1} = D^{-1}\kappa \neq 0$; if $u = A\rho_u \in \mathcal{N}$ is not parallel to $D^{-1}\kappa$ then $\rho_u \in \mathcal{K}$ and $\rho_u \not\propto \mathbf{1}$. (d) ρ and ρ' have signatures that differ by something the nuisance cannot produce exactly when $P_\perp A(\rho - \rho') \neq 0$, and the rank of $P_\perp A$ is at most the rank of $P_\perp$, which is $m - \dim \mathcal{N}$. If $m \leq \dim \mathcal{N}$ then generically $\mathcal{N} = \mathbb{R}^m$ and $P_\perp = 0$, so no pattern is identifiable. $\square$

Proposition 2. The first claim holds because each $v_k \in \mathcal{C}$ has unit norm and $\|\Pi_{\mathcal{C}}(r)\| = \sup_{v \in \mathcal{C}, \|v\| \leq 1} \langle v, r \rangle$. For the second, let $v^\star$ attain the supremum and let v_k satisfy $\angle(v_k, v^\star) \leq \gamma$, so $\|v_k - v^\star\| = 2 \sin(\gamma/2)$. Then by Cauchy–Schwarz $\langle v_k, r \rangle = \langle v^\star, r \rangle + \langle v_k - v^\star, r \rangle \geq \|\Pi_{\mathcal{C}}(r)\| - 2 \sin(\gamma/2) \|r\|$. Monotonicity is immediate since enlarging V enlarges the set over which the maximum is taken. $\square$

Proposition 3. Fix $\lambda \in [0, 1/\mu_0]$. Since $X_t \in [0, 1]$, the factor $1 + \lambda(X_t - \mu_0) \geq 1 - \lambda\mu_0 > 0$, so $W_t^\lambda \geq 0$. The bet λ is deterministic and μ_0 is an $\mathcal{F}_0$ -measurable constant, so $\mathbb{E}[W_t^\lambda \mid \mathcal{F}_{t-1}] = W_{t-1}^\lambda (1 + \lambda(\mathbb{E}[X_t \mid \mathcal{F}_{t-1}] - \mu_0)) \leq W_{t-1}^\lambda$ by (10) and $\lambda \geq 0$. Thus each W_t^λ is a non-negative supermartingale with initial value one, and so is the finite mixture W_t , convex combinations of supermartingales being supermartingales. Ville’s inequality gives $\mathbb{P}(\exists t : W_t \geq 1/\alpha) \leq \mathbb{E}[W_{t_0}] \alpha = \alpha$. Nothing in the argument uses independence across epochs, the marginal law of X_t , or the accuracy of $\hat{\beta}_{t-1}$; only predictability of the bets and of the plug-in, which holds by construction, and the conditional mean bound

(10). The same argument covers predictable bet sequences λ_t , since λ_t then factors out of the conditional expectation. $\square$

Proposition 4. (i) Under $m_t \leq \delta = \hat{\delta}$, $\mathbb{E}[1 + \lambda(X_t - \mu_0) \mid \mathcal{F}_{t-1}] = 1 + \lambda m_t \leq 1 + \lambda \delta$, so dividing the increment by the deterministic factor $1 + \lambda \delta$ restores the supermartingale property; Ville's inequality applies as in Proposition 3.

(ii) Let $\iota_s = \mathbf{1}\{s < \sigma\}$. Since m_s is $\mathcal{F}_{s-1}$ -measurable and $\hat{\delta}$ is $\mathcal{F}_{t_0}$ -measurable, $\{s < \sigma\} = \{m_u \leq \hat{\delta} \mid \forall t_0 < u \leq s\} \in \mathcal{F}_{s-1}$, so ι_s is predictable. Define $V_t^\lambda = \prod_{s=t_0+1}^t f_s$ with $f_s = \frac{1+\lambda(X_s-\mu_0)}{1+\lambda\hat{\delta}}$ if $\iota_s = 1$ and $f_s = 1$ otherwise. For $\iota_s = 1$, $\mathbb{E}[f_s \mid \mathcal{F}_{s-1}] = (1 + \lambda m_s)/(1 + \lambda \hat{\delta}) \leq 1$; for $\iota_s = 0$ it equals one. Hence V^λ , and the mixture V , are non-negative supermartingales with initial value one and $\mathbb{P}(\exists t : V_t \geq 1/\alpha) \leq \alpha$. On $\{\sigma = \infty\}$ we have $V = \widetilde{W}$, so $\mathbb{P}(\exists t : \widetilde{W}_t \geq 1/\alpha) \leq \alpha + \mathbb{P}(\sigma < \infty)$.

(iii) Put $\xi_s = X_s - \mathbb{E}[X_s \mid \mathcal{F}_{s-1}]$. Because $X_s \in [0, 1]$, conditionally on $\mathcal{F}_{s-1}$ the variable ξ_s lies in an interval of length one, so Azuma–Hoeffding (conditional Hoeffding lemma) gives $\mathbb{P}(-n^{-1} \sum_{s \in C} \xi_s > u) \leq e^{-2nu^2}$. With $u = \sqrt{\log(1/\varepsilon)/(2n)}$, with probability at least $1 - \varepsilon$ we have $\bar{m}_C = \bar{X}_C - \mu_0 - n^{-1} \sum_{s \in C} \xi_s \leq \bar{X}_C - \mu_0 + u$, hence $\bar{m}_C \leq \hat{\delta}_\varepsilon$ (the truncation at zero cannot bind against a negative upper bound on $\bar{m}_C$). On that event representativeness gives $\sup_{t > t_0} m_t \leq \bar{m}_C \leq \hat{\delta}_\varepsilon$, so $\sigma = \infty$. Thus $\mathbb{P}(\sigma < \infty) \leq \varepsilon$, and part (ii) at level $\alpha - \varepsilon$ gives $\text{FWFA} \leq (\alpha - \varepsilon) + \varepsilon = \alpha$. $\square$

Proposition 5. Write $Z_s = \log(1 + \lambda(X_s - \mu_0)) - \log(1 + \lambda \hat{\delta})$, so that $L_t^\lambda = L_{t^*}^\lambda + \sum_{s=t^*+1}^t Z_s$ for $t > t^*$. The increments are bounded: $Z_s \leq B_\lambda$ and $Z_s \geq \log(1 - \lambda \mu_0) - \log(1 + \lambda \hat{\delta}) > -\infty$ because $\lambda < 1/\mu_0$. Since the mixture dominates its components, $\widetilde{W}_t \geq \widetilde{W}_t^\lambda/|\Lambda|$, so $L_t^\lambda \geq b$ implies $\widetilde{W}_t \geq 1/\alpha$ and hence $\tau \leq T := \inf\{t > t^* : L_t^\lambda \geq b\}$. Let $e_s = \mathbb{E}[Z_s \mid \mathcal{F}_{s-1}] \geq \gamma_\lambda$ and $\xi_s = Z_s - e_s$, a bounded martingale difference sequence. For $n > t^*$, $T_n = T \wedge n$ is a bounded stopping time, so optional stopping gives

$$\mathbb{E}\left[\sum_{s=t^*+1}^{T_n} Z_s \mid \mathcal{F}_{t^*}\right] = \mathbb{E}\left[\sum_{s=t^*+1}^{T_n} e_s \mid \mathcal{F}_{t^*}\right] \geq \gamma_\lambda \mathbb{E}[T_n - t^* \mid \mathcal{F}_{t^*}].$$

If $L_{t^*}^\lambda \geq b$ then $T = t^*$ and there is nothing to prove; otherwise, before T the log-wealth is below b , and at T it exceeds b by at most B_λ , so $L_{T_n}^\lambda \leq b + B_\lambda$ and $\sum_{s=t^*+1}^{T_n} Z_s = L_{T_n}^\lambda - L_{t^*}^\lambda \leq b + B_\lambda + (L_{t^*}^\lambda)^-$. Combining, $\gamma_\lambda \mathbb{E}[T_n - t^* \mid \mathcal{F}_{t^*}] \leq b + B_\lambda + (L_{t^*}^\lambda)^-$; letting $n \rightarrow \infty$ (monotone convergence) bounds $\mathbb{E}[T - t^* \mid \mathcal{F}_{t^*}]$, and $(\tau - t^*)^+ \leq T - t^*$ gives (13). The condition $\gamma_\lambda > 0$ is exactly the requirement that the damage generate evidence faster than the allowance consumes it. $\square$

Proposition 6. For $t \leq s_k$ the component $\widetilde{W}^{(k)}$ is constant at one, and for $t > s_k$ it is, by the argument of Proposition 4(i) applied from s_k with the conditioning $\mathcal{F}_{s_k}$, a non-negative supermartingale. A sum with weights π_k of non-negative supermartingales (monotone convergence for infinitely many terms) is a non-negative supermartingale, with initial value $\sum_k \pi_k \leq 1$, and Ville's inequality gives $\mathbb{P}(\exists t : W_t^R \geq 1/\alpha) \leq \alpha \sum_k \pi_k \leq \alpha$. For the delay, $W_t^R \geq \pi_{k^*} \widetilde{W}_t^{(k^*)}$, so an alarm occurs no later than the first time the log-wealth of component k^* , mixed over Λ , reaches $\log(1/\alpha) + \log(1/\pi_{k^*})$, and the proof of Proposition 5 applies with b replaced by $b + \log(1/\pi_{k^*})$. $\square$